

Classical Mechanics and Dynamical Systems

Lectures 13–29 — Prof. V. Balakrishnan

Handwritten lecture notes, transcribed

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1 Lecture 13: Dynamical Symmetry – 1

Understanding the relation of symmetry $\Leftrightarrow$ invariance $\Rightarrow$ conservation laws. First step in this regard was taken by Emmy Noether.

(conserved quantities $\vec{P}$, $\vec{L}$, $E(H) \dots$)

Noether's theorem :-

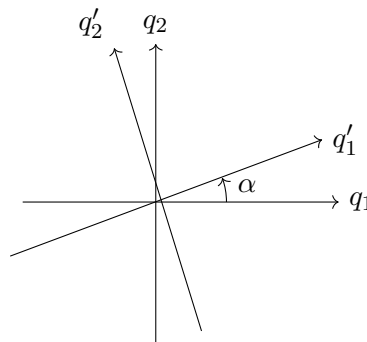
Let us take a case of a particle moving in 2-D :-

$$H(q_1, q_2, p_1, p_2) = \frac{\vec{p}^2}{2m} + V(q_1, q_2)$$

$$p^2 = p_1^2 + p_2^2$$

If the potential has spherical symmetry (in 2-D its rotational) then we know that H is C.O.M.

$$\text{if } V(q_1, q_2) = f(r), \quad r = \sqrt{q_1^2 + q_2^2}$$



In rotated frame of reference what remains constant :-

- 1) H
- 2) EOM don't change
- 3) Solution space/set do not change

Q) Does solution remains same/const?

Lets take simplest case

$$x^2 = 4$$

$$x^2 - 4 = 0$$

$$(x - 2)(x + 2) = 0$$

$$x = 2, -2 \quad \{-2, 2\}$$

after rota transformation

$$x'^2 = 4 \quad x' = -x$$

$$(x' - 2)(x' + 2) = 0$$

$$x' = -2, +2 \quad \{-2, 2\}$$

since $x' = -x$

we deduce that the +2 root changed to -2 and $-2 \rightarrow +2$,

but set of solution remains constant/same

even though the elements of set of solution can convert into one another.

So the defⁿ of dynamical symmetry becomes,

* Dynamical symmetry :-

The dynamical symmetry of a given system is a set of transformations of phase space variables, such that EOM don't change and as a consequence, the set of solutions do not change.

(1) Understanding Dynamical symmetry in Lagrangian formalism :-

$$\begin{array}{ccc}
 L(q, \dot{q}) & \longrightarrow & L'(Q(\alpha), \dot{Q}(\alpha)) \quad \alpha = \text{transformation parameter} \\
 \uparrow & & \downarrow \\
 \text{(Autonomous dyn. system)} & & L' \quad (L' = L \text{ if } \alpha = 0 \text{ "by def"}^n) \\
 & & Q(0) = q
 \end{array}$$

EOM do not change if $L' = L$

or, Invariance $\Rightarrow L' = L$

$$\begin{aligned}
 \frac{dL(Q(\alpha), \dot{Q}(\alpha))}{d\alpha} &= 0 \\
 \Rightarrow \frac{\partial L}{\partial Q} \frac{dQ}{d\alpha} + \frac{\partial L}{\partial \dot{Q}} \frac{d\dot{Q}}{d\alpha} &= 0 \\
 \Rightarrow \frac{d}{dt} \left(\frac{\partial L}{\partial \dot{Q}} \right) \frac{dQ}{d\alpha} + \frac{\partial L}{\partial \dot{Q}} \frac{d}{d\alpha} \left(\frac{dQ}{dt} \right) &= 0
 \end{aligned}$$

α, t are independent so

$$\begin{aligned}
 \frac{d}{dt} \left(\frac{\partial L}{\partial \dot{Q}} \right) \frac{dQ}{d\alpha} + \frac{\partial L}{\partial \dot{Q}} \frac{d}{dt} \left(\frac{dQ}{d\alpha} \right) &= 0 \\
 \Rightarrow \frac{d}{dt} \left(\left(\frac{\partial L}{\partial \dot{Q}} \right) \cdot \left(\frac{dQ}{d\alpha} \right) \right) &= 0 \\
 \Rightarrow \frac{\partial L}{\partial \dot{Q}} \left(\frac{dQ}{d\alpha} \right) &= \text{const of motion}
 \end{aligned}$$

in our frame

$$\frac{\partial L}{\partial \dot{Q}} \left(\frac{dQ}{d\alpha} \right) \equiv \frac{\partial L}{\partial \dot{q}} \left(\frac{dQ}{d\alpha} \right) \Big|_{\alpha=0} = p \frac{dQ}{d\alpha} \Big|_{\alpha=0} = \underline{\text{COM}}$$

Remember, that Lagrangian (L) is not unique, we can add total time derivative of some fun. of q, t , therefore it is not necessary that

$$\text{Invariance} \Rightarrow \underline{L' = L}$$

we can have $L' \neq L$ still EOM don't change.

Note :- The crucial point of Noether's theorem is that set of transformations should be continuous, that's why we differentiated with 'α', and it must be connected with Identity (original $(q, \dot{q})$.) In other words if we have original coordinates and then we make a transformation continuously from no transformation at all.

Parity is not continuous transformation

Continuous Transformation

- Rotation
- shift in origin
- shift in origin in time
- shear, scale
- gauge transformations (vary χ continuously)

↓
(gauge invariance $\Rightarrow$ leads to conservation of charge).

Symmetry :- If Lagrangian is invariant under continuous set of transformations then Noether's theorem tells us $\exists$ a conserved quantity for each transformation parameter.

eg: In shift of origin we $(x, y, z) \longrightarrow (x + \alpha, y + \beta, z + \gamma)$

we have 3 parameters of transformation i.e. α, β, γ

So according to Noethers theorem we will have 3 conserved quantity.

(2) Understanding in Hamiltonian Framework :-

$$H(q, p) : \quad \dot{q}_i = \frac{\partial H}{\partial p_i}, \quad \dot{p}_i = -\frac{\partial H}{\partial q_i} \quad \boxed{\dot{\tilde{X}} = J \nabla H}$$

$\uparrow$ (set of EOM)

$$\exists \quad \left(\begin{array}{l} \{q_i, q_j\} = 0 = \{p_i, p_j\} \\ \{q_i, p_j\} = \delta_{ij} \end{array} \right)$$

Q.) What set of transformations leaves Hamiltons EOM unchanged

Ans) canonical transformations.

$$\dot{Q}_i = \frac{\partial K}{\partial P_i}, \quad \dot{P}_i = -\frac{\partial K}{\partial Q_i} \quad \boxed{\dot{\tilde{\xi}} = J \nabla_{\xi} K}$$

$\uparrow$ (EOM)

also

$$\begin{aligned} \{Q_i, P_j\} &= \delta_{ij} \\ \{Q_i, Q_j\} &= 0 = \{P_i, P_j\} \end{aligned}$$

to see set of solutions remain unchanged, we need only those canonical transformations for which $\underline{K = H}$, else it won't be a symmetry. (How?)

Suppose $\tilde{X} = (q, p) \xrightarrow{\text{C.T.}} \tilde{\xi} = (Q, P)$

where $\underline{X}, \underline{\xi}$ are set of variables.

$$\left. \begin{aligned} \{Q_i, P_j\} &= \delta_{ij} = \sum_{k=1}^n \left(\frac{\partial Q_i}{\partial q_k} \frac{\partial P_j}{\partial p_k} - \frac{\partial Q_i}{\partial p_k} \frac{\partial P_j}{\partial q_k} \right) \\ \{Q_i, Q_j\} &= 0 = \sum_{k=1}^n \left(\frac{\partial Q_i}{\partial q_k} \frac{\partial Q_j}{\partial p_k} - \frac{\partial Q_i}{\partial p_k} \frac{\partial Q_j}{\partial q_k} \right) \\ \{P_i, P_j\} &= 0 = \sum_{k=1}^n \left(\frac{\partial P_i}{\partial q_k} \frac{\partial P_j}{\partial p_k} - \frac{\partial P_i}{\partial p_k} \frac{\partial P_j}{\partial q_k} \right) \end{aligned} \right\} \Rightarrow \left(\frac{\partial \underline{\xi}}{\partial \underline{x}} \right)^T J \left(\frac{\partial \underline{\xi}}{\partial \underline{x}} \right) = J$$

all these relations must be valid to qualify for transformation to be canonical.

$$J = \begin{pmatrix} O_n & I_n \\ -I_n & O_n \end{pmatrix} \quad I_n, O_n \text{ are } n \times n \text{ matrices.}$$

any $\{A, B\}$ can be written as, $\{A, B\} = (\nabla A)^T J (\nabla B) \quad (M_{p \times q})^T J_{m \times m} M_{p \times q}, q = 2n, p = 2n$

If a matrix (M) follows

$$\text{if, } M^T J M = J,$$

then M is called symplectic matrix

Recall,

$$J^2 = -I, \quad J^T = J^{-1} = -J$$

so symplectic matrix have inverse so canonical transformations Matrix is invertible. $M = \left(\frac{\partial \underline{\xi}}{\partial \underline{x}} \right) = \text{non-singular } \underline{|M|} \neq 0$

$$\Updownarrow$$

we know, canonical transformation must be inverse transformations,

also, Product of two symplectic matrix is also a symplectic matrix. (as set of symplectic matrices form a group).

$$M_1.M_2 = M_3 \quad M_1, M_2, M_3 \rightarrow \text{symplectic matrices.}$$

physically it implies that two canonical transformations applied serielly are equivalent to a single canonical transformation.

Note:- The inverse, composition law suggests set of canonical transformation for n-d.o.f Hamiltonian system form a group. It's the group of $2n \times 2n$ symplectic matrices. Called

$$Sp(2n, \mathbb{R}).$$

$\hookrightarrow$ Group of C.T.

Note: J is also a symplectic matrix

$$M^T J M = J$$

$$\text{LHS, } \Rightarrow J^T J J \Rightarrow (-J) J^2 = (-J)(-I) = J = \underline{\text{RHS}}$$

Q) what is C.T. corresponding Symplectic matrix 'J' ?

$$\left. \begin{matrix} Q = -p \\ P = q \end{matrix} \right\} \Leftrightarrow (M = J)$$

Note:- We know that group of orthogonal matrices in 3-dimensions have three parameters, (for rotation)

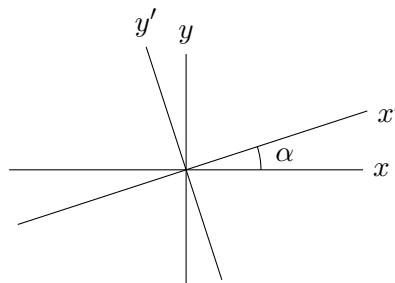
$$SO(n)$$

$\hookrightarrow$ Group of orthogonal matrices in n-dimension with determinant (1)

(Digression:)

Q) How many parameters for n-D ?

Ans: (a) rotation in 2-D about point (origin).



$$SO(2); \begin{pmatrix} \cos \alpha & \sin \alpha \\ -\sin \alpha & \cos \alpha \end{pmatrix}$$

$\downarrow$

orthogonal.

set of transformation is homogeneous (origin unchanged).

2-D rotation is (linear, Homogeneous)

$$(|SO(2)| = \pm 1, \text{ length unchanged.})$$

Defining rotation :-

(for n-dimensions)

- linear
- Homogeneous
- distances unchanged $\Rightarrow$ Matrice should be orthogonal
- $|SO(n)| = \pm 1$ (unimodular)

Note:- In 3-D if we rotate by a single axis, it is not termed as rotation. as from defⁿ we have not defined rotation as rotation about a single axis. For Rotation T should be linear, Homogeneous, orthogonal, and unimodular.

Note:- If the transformation is homogeneous (origin – fixed) to achieve distances unchanged the transformation matrix should be orthogonal. and if we want orientation to be unchanged we

need it to be unimodular. $|M| = 1$. So as we know $\det(\text{orthogonal } m) = \pm 1$. we choose $\underline{+1}$ in rotation. for reflection $\underline{|M| = -1}$.

In 3-D we can do rotation in a plane, where two variables are affected at a time. For two variables affected at a time; no. of rotations are nC_2 . for n-dimension.

$$\text{no. of rotation} \Rightarrow {}^nC_2 = \frac{n(n-1)}{2}$$

$$\text{for } n = 3, \quad \text{no. of rotations} = \frac{3 \cdot 2}{2} = 3$$

So we think (mistakenly) that rotation is occurring about a axis, but really we mean is rotation in xz plane, yz plane, xy plane.

$$\text{Hence, only for 3-D } \left(\frac{n(n-1)}{2} = n \right)$$

So $SO(n)$ has nC_2 generators.

In exactly the same way,

Now we can ask how many generators does this symplectic group $Sp(2n, R)$ have? (or) How many possible parameters we need to specify all possible canonical transformations.

let us make Infinitesimal transformation.

$$M = I + \epsilon G_{(2n \times 2n)} \longrightarrow \text{Generator of Transformation matrix} \\ \longleftrightarrow \text{parameter}$$

$$\begin{aligned} M^T J M &= J \\ \Rightarrow (I + \epsilon G)^T J (I + \epsilon G) &= J \\ \Rightarrow (I + \epsilon G^T) J (I + \epsilon G) &= J \\ \Rightarrow (J + \epsilon G^T J) (I + \epsilon G) &= J \\ \Rightarrow J + \epsilon G^T J + \epsilon J G + \epsilon^2 G^T J G &= J \quad \text{dropping } \underline{\underline{\epsilon^2}} \\ \Rightarrow J + \epsilon G^T J + \epsilon J G &= J \\ \Rightarrow \boxed{G^T J = -J G} \\ \Rightarrow G^T &= -J G J^{-1} \quad (J^{-1} = -J) \\ \Rightarrow \boxed{G^T = J G J} \end{aligned}$$

So Generator of symplectic transformation must satisfy above equation.

so symplectic matrices are generated by these matrices which satisfies $G^T = J G J$

exercise $\rightarrow$ Show that $Sp(2n)$ has $2(2n+1)$ generators.

what are generators? The way we look at it can be interpreted by looking at infinitesimal transformation.

$$\begin{aligned} SO(2) &= \begin{pmatrix} \cos \alpha & \sin \alpha \\ -\sin \alpha & \cos \alpha \end{pmatrix} \xrightarrow{(\alpha \rightarrow \delta \alpha)} \begin{pmatrix} 1 & \delta \alpha \\ -\delta \alpha & 1 \end{pmatrix} \\ &= I + \begin{pmatrix} 0 & \delta \alpha \\ -\delta \alpha & 0 \end{pmatrix} \\ &= I + \delta \alpha \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix} \end{aligned}$$

$\hookrightarrow$ Generator.

$\hookrightarrow$ parameter

for $SO(3)$ we will have such 3 generators.

and for $Sp(2n)$ we will have $2(2n + 1)$ generators.

Note:- we actually like to write generators as Hermitian matrix. because we need most of time to exponentiate hermitian matrix. as, e^{iH} = unitary matrix.

Eg 2-D oscillator :-

$$H = \frac{1}{2} (q_1^2 + p_1^2 + w^2 q_2^2 + p_2^2)$$

quasi periodic ($w = \text{irrational}$ [?])

periodic but complicated ($w = \text{rational}$ [?])

$\downarrow$

($w = \text{rational no}$)

if ($w = 1$)

$$H = \frac{1}{2} (q_1^2 + p_1^2 + q_2^2 + p_2^2)$$

isotropic (same k in all directio [?])

$\hookrightarrow$ spring constant [?]

H is invariant under $SO(4)$.

$\hookrightarrow$ set of rotations in 4-D phase space.

Note:- we could have dynamical symmetry arising as a consequence of transformations not just from of potentials (not just of physical coordinate) but it could also involve the momenta (phase-space).

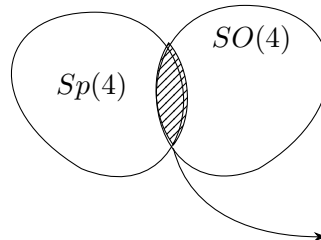
$\rightarrow$ no. of generators $SO(4) = {}^4C_2 = 6$ -(1)

$\rightarrow$ set of canonical transformations ($n = 2$) $\rightarrow Sp(2 \times 2)$

$Sp(4)$, has $2(4 + 1) = 10$ generators. -(2)

From (1) and (2) we deduce that all transformations which leaves hamiltonian unchanged need not be canonical transformations, similarly all C.T need not leave 'H' unchanged.

For symmetry we need transformation to be canonical and which leaves 'H' unchanged.
($K = H$)



Dynamical symmetry group is

$$Sp(4) \cap SO(4). \sim SU(2)$$

$\hookrightarrow$ Isomorphic

$SU(2)$ set of 2×2 matrices which are unitary and unimodular ($|M| = 1$)

$$U \text{ is unitary} \Rightarrow \boxed{U^\dagger U = I}$$

Q. Is it true $U^\dagger U = I = U U^\dagger$

Ans, Only true for finite dimensional matrices (as left inverse becomes equal to right inverse). But for infinitesimal matrices $U^\dagger U = I \not\Rightarrow U U^\dagger$, But we are concerned with finite-dimensional matrices, so it does not matter.

$$\begin{aligned} U^\dagger U = I &\Rightarrow \text{partial isometry} \\ U U^\dagger = I &\Rightarrow \quad \quad \quad \text{“} \quad \quad \text{”} \end{aligned}$$

verify that following J_1, J_2 are c.o.m are true.

$$J_2 = \frac{1}{2} (q_1 q_2 + p_1 p_2) \quad J_1 = \frac{1}{4} (q_1^2 + p_1^2 - q_2^2 - p_2^2)$$

$$J_3 = \frac{1}{2} (q_1 p_2 - q_2 p_1)$$

verify, $(\{J_i, H\} = 0, \quad \{J_i, J_j\} = \epsilon_{ijk} J_k)$.

2 Lecture 14: (Finding dyn. symmetry group $SU(n)$)

Recall 2-D isotropic oscillator :-

$$\begin{pmatrix} m = 1 \\ w = 1 \end{pmatrix}$$

$$H = \frac{1}{2} (q_1^2 + p_1^2 + q_2^2 + p_2^2)$$

$$F_1 = H_1 \quad (q_1^2 + p_1^2)$$

$$F_2 = H_2 \quad (q_2^2 + p_2^2)$$

its 2-dof and we have 2 c.o.M which are in involution with each other ($\{F_1, F_2\} = 0$. (Integrable)

Note:-

The dynamical symmetry will be that set of transformations of 4 phase space variables which leaves the H, EOM unchanged and hence preserves the solution set.

$Sp(2n) = Sp(4)$ = Group of canonical Transformations for this H.

$SO(4) = SO(4)$ = Group of rotations which ‘leaves’ ‘H’ unchanged

$SU(n) = SU(2)$ = Group of C.T. which leaves ‘H’ unchanged.

(also called as the dyn. symmetry of this H).

Let's focus on $SU(2)$. :-

$SU(2)$ consists of all 2×2 matrices, $M = \begin{pmatrix} a & b \\ c & d \end{pmatrix}$, such that

($\Rightarrow$ implies)

$$\begin{aligned} M^\dagger M &= I \\ \det(M) &= +1 \end{aligned}$$

There are $4 \times 2 = 8$ parameters as each entry can be complex no.

$$\begin{aligned} a &= a_1 + ia_2 \\ b &= b_1 + ib_2 \\ c &= c_1 + ic_2 \\ d &= d_1 + id_2 \end{aligned} \quad (a_1, a_2, b_1, b_2, c_1, c_2, d_1, d_2)$$

$\hookrightarrow 8$ parameters.

using, $M^\dagger M = I$ or $M^{-1} = M^\dagger$

$\rightarrow$ we will have 4 conditions (one for each element).

Therefore no. of parameters are reduced to 4.

after imposing $|M| = +1$, one more parameter is lost— so total of three parameters are left.

(Digression) :-

$$M^\dagger = M^{-1} \quad M = \begin{pmatrix} \alpha & \beta \\ \gamma & \delta \end{pmatrix}$$

$$M^\dagger = \begin{pmatrix} \alpha^* & \gamma^* \\ \beta^* & \delta^* \end{pmatrix} \quad \text{--- (1)}$$

$$M^{-1} = \frac{1}{|M|} \begin{pmatrix} \delta & -\gamma \\ -\beta & \alpha \end{pmatrix}^T = \frac{1}{(|M|)} \begin{pmatrix} \delta & -\beta \\ -\gamma & \alpha \end{pmatrix} \quad \text{--- (2)}$$

equating (1) and (2) and $|M| = \alpha\delta - \beta\gamma = 1$

$$\alpha^* = \delta \quad \text{--- (3)}$$

$$\beta^* = -\gamma \quad \text{--- (4)}$$

$$\gamma^* = -\beta \quad \text{--- (5)}$$

$$\delta^* = \alpha \quad \text{--- (6)}$$

$$\alpha^* = \delta, \delta^* = \alpha \Rightarrow \begin{aligned} (\alpha_1 + i\alpha_2)^* &= (\delta_1 + i\delta_2) \Rightarrow \alpha_1 = \delta_1 \\ (\delta_1 + i\delta_2)^* &= \alpha_1 + i\alpha_2 \Rightarrow \alpha_2 = -\delta_2 \end{aligned}$$

so M becomes

$$M = \begin{pmatrix} \alpha & \beta \\ -\beta^* & \alpha^* \end{pmatrix}, \quad \begin{aligned} |M| &= 1 \\ \alpha\delta - \beta\gamma &= 1 \\ \alpha\alpha^* + \beta\beta^* &= 1 \\ |\alpha|^2 + |\beta|^2 &= 1 \end{aligned}$$

Hence, any (2×2) unimodular, unitary matrix can be written is

$$\begin{pmatrix} \alpha & \beta \\ -\beta^* & \alpha^* \end{pmatrix} \quad \text{with,} \quad (|\alpha|^2 + |\beta|^2 = 1) \quad \begin{cases} \alpha = \alpha_1 + i\alpha_2 \\ \beta = \beta_1 + i\beta_2 \end{cases}$$

The Generators of $SU(2)$ are c.o.M which are not in involution with each other. and turns out these are J_1, J_2, J_3

$$\begin{aligned} J_1 &= \frac{1}{4} (q_1^2 + p_1^2 - q_2^2 - p_2^2) \\ J_2 &= \frac{1}{2} (q_1 q_2 + p_1 p_2) \\ J_3 &= \frac{1}{2} (q_1 p_2 - p_1 q_2) \end{aligned} \quad \left. \begin{aligned} \{J_1, J_2\} &= J_3 \\ \{J_2, J_3\} &= J_1 \\ \{J_3, J_1\} &= J_2 \end{aligned} \right\} \text{ (or) } \begin{cases} \{J_i, J_j\} \\ = \epsilon_{ijk} J_k \end{cases}$$

(so 3 parameters require 3 generators)

Q) How many independent parameters in $(n \times n)$ unitary matrix

Ans, $2n^2 - n^2 = n^2$

So, for $SU(n)$ we have $(n^2 - 1)$ independent parameters where 1 parameter got reduced from $|M| = \pm 1$ condition.

Therefore, i.e. ($SU(3)$ has $3^2 - 1 = 8$ generators)

For 3-D Isotropic oscillator :-

$Sp(2n)$. $Sp(6)$ is group of C.T.

$SO(N)$ $SO(6)$ is group of symmetry transformations

dimension of phasespace

$SU(8)$ is one required dynamical symmetry group.

The Kepler problem :-

$$H(\vec{r}, \vec{p}) = \frac{\vec{p}^2}{2m} + \left(-\frac{k}{r}\right) \quad \dots \quad \left(\begin{array}{l} \text{assuming interaction} \\ \text{is attractive.} \end{array} \right)$$

Note:-

$$V(r) = \begin{cases} -k/r \\ r^2 \end{cases} \quad \text{are the only potentials for which bounded orbits are closed curves.}$$

↓

$$\left(\begin{array}{l} \text{periodic} \\ \text{orbits} \end{array} \right).$$

$$V(r) = r^2 \quad \text{3-D oscillator, with } SU(3)$$

Q) what is symmetry group of Kepler's hamiltonian?

C.O.M :- $H, \vec{L}^2, \vec{L} \cdot \hat{n} \rightarrow 3$ independent c.o.m which are in involution with each other.

$$\vec{L} \text{ is c.o.m but } \{L_i, L_j\} = \epsilon_{ijk} L_k \quad \left(\begin{array}{l} \text{not in involution} \\ \text{with each other} \end{array} \right).$$

Note

Any central potential problem will have these c.o.m but there is something special with kepler problem as there exist further C.O.M ($\vec{A}$).

$$\vec{A} = \vec{p} \times \vec{L} - \frac{mk\vec{r}}{r} \quad \left(\begin{array}{l} A = f(L) \text{ so its not} \\ \text{independent} \end{array} \right)$$

(laplace- Runge -lenz vector)

(verifying $\vec{A}$ is C.O.M) :-

($\vec{L}$ is c.o.m)

$$\frac{d\vec{A}}{dt} = \frac{d\vec{p}}{dt} \times \vec{L} + \vec{p} \times \frac{d\vec{L}}{dt} + \frac{mk}{r^2} \frac{dr}{dt} \vec{r} - \frac{mk}{r} \frac{d\vec{r}}{dt}$$

$$\begin{aligned}
&\text{for } V = -k/r \\
&F = -\nabla V \\
&= -\nabla(-k/r) \\
&= \nabla(k/r) \\
&= -k/r^2 \hat{e}_r \\
&\frac{d\vec{p}}{dt} = F = \frac{-k}{r^3} \vec{r}
\end{aligned}$$

$$\begin{aligned}
\frac{d\vec{A}}{dt} &= \frac{-k}{r^3} \vec{r} \times (\vec{r} \times \vec{p}) + \frac{mk}{r^2} \frac{dr}{dt} \vec{r} - \frac{mk}{r} \frac{d\vec{r}}{dt} \\
\frac{d\vec{A}}{dt} &= \frac{-k}{r^3} (\vec{r} \times (\vec{r} \times \vec{p})) + \frac{mk}{r^2} \frac{dr}{dt} \vec{r} - \frac{k\vec{p}}{r} \quad \left(\vec{p} = m \frac{d\vec{r}}{dt} \right)
\end{aligned}$$

$$\begin{aligned}
\bar{a} \times (\bar{b} \times \bar{c}) &= \text{vector in plane of } \bar{b}, \bar{c}. \\
&= \alpha \vec{b} + \beta \vec{c}
\end{aligned}$$

LHS is linear, α, β are scalars formed out of a, b, c as LHS is linear RHS must be linear,

$$\begin{aligned}
&\alpha = \lambda(\vec{a} \cdot \vec{c}) \\
\Rightarrow &\beta = \mu(\vec{a} \cdot \vec{b})
\end{aligned}$$

‘ λ & μ ’ should be universal, it can’t depend on $\vec{A}, \vec{B}, \vec{C}$.

$$\bar{a} \times (\bar{b} \times \bar{c}) = \lambda(\bar{a} \cdot \bar{c}) \vec{b} + \mu(\bar{a} \cdot \bar{b}) \vec{c}$$

interchang b, c .

$$\vec{a} \times (\vec{c} \times \vec{b}) = -\bar{a} \times (\bar{b} \times \bar{c})$$

only way it can happen is, $\mu = -\lambda$

$$\bar{a} \times (\bar{b} \times \bar{c}) = \lambda \left((\bar{a} \cdot \bar{c}) \vec{b} - (\bar{a} \cdot \bar{b}) \vec{c} \right)$$

turns out $\lambda = 1$ using special case $\hat{i} \times (\hat{i} \times \hat{j}) = \hat{i} \times \hat{k} = -\hat{j}$

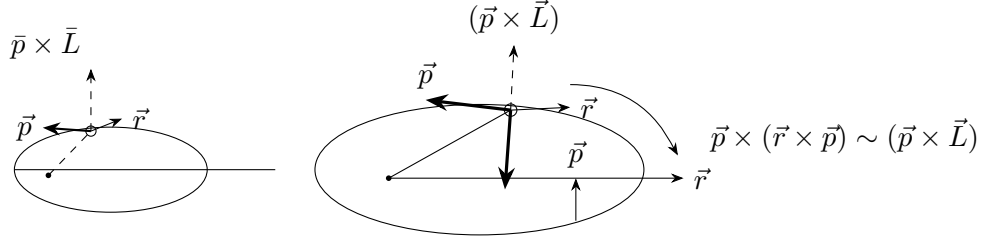
$$\lambda \{ (\hat{i} \cdot \hat{j}) \hat{i} - \hat{j}(\hat{i} \cdot \hat{i}) \} = -\hat{j} \Rightarrow \boxed{\lambda = 1}$$

$$\begin{aligned}
\frac{d\vec{A}}{dt} &= \frac{-k}{r^3} [(\vec{r} \cdot \vec{p}) \vec{r} - r^2 \vec{p}] + \frac{mk}{r^2} \frac{dr}{dt} \vec{r} - \frac{k\vec{p}}{r} \\
&= \frac{-k}{r^3} (\vec{r} \cdot \vec{p}) \vec{r} + \frac{k\vec{p}}{r} + \frac{mk}{r^2} \frac{dr}{dt} \vec{r} - \frac{k\vec{p}}{r} \\
&= \frac{-k(m)}{r^3} \left(\vec{r} \cdot \frac{d\vec{r}}{dt} \right) \vec{r} + \frac{mk}{r^2} \frac{dr}{dt} \vec{r} \quad \left(\vec{u} \cdot \frac{d\vec{u}}{dt} = u \frac{du}{dt} \right) \\
&= \frac{-mk}{r^3} r \frac{dr}{dt} \vec{r} + \frac{mk}{r^2} \frac{dr}{dt} \vec{r}
\end{aligned}$$

$$\frac{d\vec{A}}{dt} = 0 \quad \text{therefore } \vec{A} \text{ is C.O.M.}$$

$\vec{A}$ is c.o.m therefore it remains unchanged in direction and magnitude.

$$\vec{A} = \vec{p} \times (\vec{r} \times \vec{p}) - \frac{mk\vec{r}}{r}$$



(dir. of $\vec{A}$ lies along semi major axis.)

The $\frac{d\vec{A}}{dt} = 0$, implies orbit does not show precession.

$$\left[\begin{array}{l} \text{Binary stars precess } \simeq 4^\circ \text{ per orbit} \\ \text{(Precession of mercury } \simeq 500'' \text{ per century)} \end{array} \right]$$

So in total we got $H, \vec{L}, \vec{A}$ total of $(1+3+3) = 7$ C.O.M but phase space is only 6-dimensional. It turns out that they are not independent of each other. $\left(\begin{array}{l} \vec{A}^2 \text{ is expressed in terms of } H \text{ \& } \vec{L}^2 \\ \vec{A} \cdot \vec{L} = 0. \end{array} \right.$

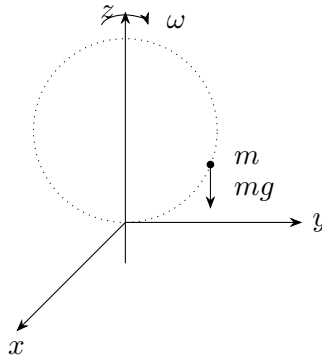
Note:- $\frac{d\vec{A}}{dt} = 0$ independent of sign of 'k' so $\exists \vec{A}$ for repulsive forces also.

* The dynamical symmetry group is $SU(4)$. instead of $SU(3)$ This extra symmetry is seen in terms of degeneracy in quantum mechanics.

This extra symmetry is because of $\vec{A}$.

Problem 1)

Beed on a rotating hoop of wire.



$$V = \begin{cases} 0 & \text{at } z = 0 \\ mgz & z = z \end{cases}$$

$$L = T - V$$

$$L = \frac{1}{2}m(\dot{x}^2 + \dot{y}^2 + \dot{z}^2) - mgz$$

constraint $\longrightarrow$

$$y^2 + (z - R)^2 = R^2$$

The moment we provided $\vec{w}$ to hoop we need cylindrical symmetry. (s, ϕ, z)

$$\begin{aligned} x &= s \cos \phi, & \dot{x} &= \dot{s} \cos \phi - s \sin \phi \dot{\phi} \\ y &= s \sin \phi, & \dot{y} &= \dot{s} \sin \phi + s \cos \phi \dot{\phi} \\ z &= z, & \dot{z} &= \dot{z} \end{aligned}$$

so L becomes;

$$L = \frac{1}{2}m \left[\dot{s}^2 + s^2 \dot{\phi}^2 + \dot{z}^2 \right] - mgz \quad \text{--- (1)}$$

constraint becomes

$$\begin{aligned} (x^2 + y^2) + (z - R)^2 &= R^2 \\ \Rightarrow s^2 + (z - R)^2 &= R^2 \quad \text{--- (2)} \end{aligned}$$

$$\text{also, } \dot{\phi} = \omega \quad \text{--- (3)}$$

eqⁿ (2) implies, $z = R \pm \sqrt{R^2 - s^2}$

we should choose $z = R - \sqrt{R^2 - s^2}$

as for $s = 0, z \rightarrow 0$

$$\dot{z} = \frac{+(-2s)\dot{s}}{2\sqrt{R^2 - s^2}} = \frac{s\dot{s}}{\sqrt{R^2 - s^2}} \quad \text{--- (4)}$$

using eqn (3) (4) in (1). 'L' becomes

$$\begin{aligned} L &= \frac{m}{2} \left[\dot{\rho}^2 + \rho^2 \omega^2 + \frac{\rho^2 \dot{\rho}^2}{R^2 - \rho^2} \right] - mg \left(R - \sqrt{R^2 - \rho^2} \right) \\ &= \frac{m}{2} \left[\frac{\dot{\rho}^2 R^2 - \dot{\rho}^2 \rho^2 + \rho^2 \omega^2 R^2 - \rho^2 \omega^2 \rho^2 + \rho^2 \dot{\rho}^2}{R^2 - \rho^2} \right] - mgR + mg\sqrt{R^2 - \rho^2} \\ L(\rho, \dot{\rho}) &= \frac{m}{2} \left(\frac{\dot{\rho}^2 R^2}{R^2 - \rho^2} \right) + \frac{1}{2}m\rho^2 \omega^2 - mgR + mg\sqrt{R^2 - \rho^2} \end{aligned}$$

since L only depends on one dynamical variable of space (ρ) it has only one degree of freedom (ρ).

$$\boxed{\frac{\partial L}{\partial \rho} = \frac{d}{dt} \left(\frac{\partial L}{\partial \dot{\rho}} \right)}$$

$$p_\rho = \frac{\partial L}{\partial \dot{\rho}} = \frac{mR^2 \dot{\rho}}{(R^2 - \rho^2)} \quad \Rightarrow \quad \dot{\rho} = \frac{p(R^2 - \rho^2)}{mR^2}$$

$$\begin{aligned} H(\rho, p) &= p\dot{\rho} - L \\ &= p\dot{\rho} \frac{mR^2}{(R^2 - \rho^2)} - \frac{p^2(R^2 - \rho^2)^2}{m^2 R^4} - L \end{aligned}$$

$$H(\rho, p) = \frac{p^2(R^2 - \rho^2)}{mR^2} - \frac{1}{2}m\omega^2\rho^2 + mgR - mg\sqrt{R^2 - \rho^2}$$

$$\dot{\rho} = \frac{\partial H}{\partial p} = \frac{2p(R^2 - \rho^2)}{mR^2}$$

$$\dot{p} = \frac{-\partial H}{\partial \rho} = \frac{+2p^2\rho}{mR^2} + m\omega^2\rho - \frac{mg\rho}{\sqrt{R^2 - \rho^2}}$$

3 Lecture 15: (Randomness in phase-space)

How do we find eqn of phase-trajectory using Hamiltons. eqn of motion.?

$$H(q, p)$$

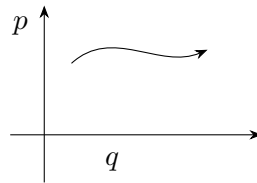
$$\dot{q}_i = \frac{\partial H}{\partial p_i}$$

$$\dot{p}_i = -\frac{\partial H}{\partial q_i}$$

Ans, we have to solve these equations; if these eqn are integrable.

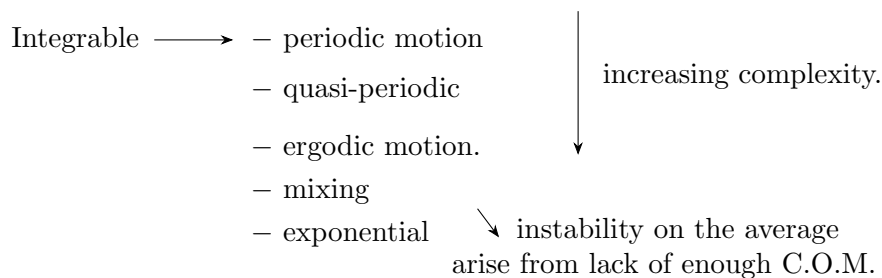
for 1-D.O.f, C.O.M = H

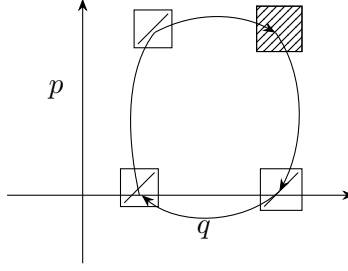
so trajectory is just given by $H = \text{const.}$



for n -D.O.f, we need n -C.O.M and find the C.T. which takes to (I, θ) variables and task is not easy to do.

(If system is integrable, we can predict what will happen at any time 't' in future provided initial 'phase-space point'.)



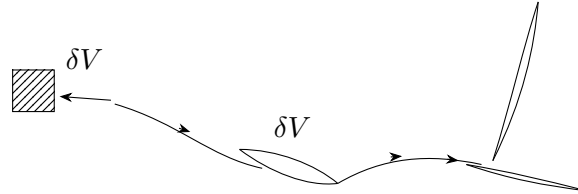


final *pos* phase space volume is closer to initial one

If initial volume element in phase space visits the neighbourhood of every point of part of phase-space, in given sufficient time, then we say motion is ergodic.

Note :-

Worse can happen to initial volume element, even when volume is preserved.

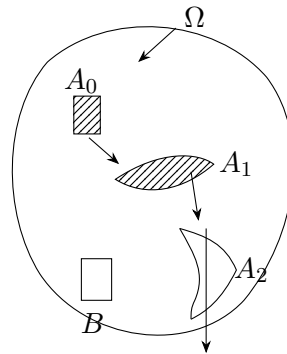


as time goes on such pieces of volume element can go as far as size of system (arbitrary far) such kind of motion is ergodic too but there is a special name to the motion called as mixing.

Mixing :-

Suppose the volume of full phase space is Ω .

A_0 — initial phase space volume



$\mu \rightarrow$ measure

If A_0 is completely mixed up, how much of A_0 is present in B .

$$\frac{\mu(A_n \cap B)}{\mu(A_0)} = \frac{\mu(B)}{\mu(\Omega)}$$

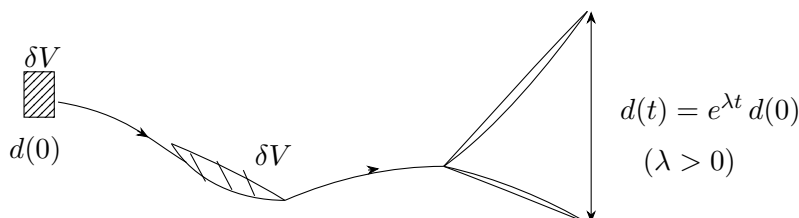
if, $\lim_{n \rightarrow \infty} \mu(A_n \cap B) = \frac{\mu(A_0) \mu(B)}{\mu(\Omega)}$

then the dynamics is said to be strongly mixing.

Note :- mixing implies ergodicity whereas ergodicity does not imply mixing as we can have ergodic motion without mixing (distortion of phase-space element).

• Rate of mixing :-

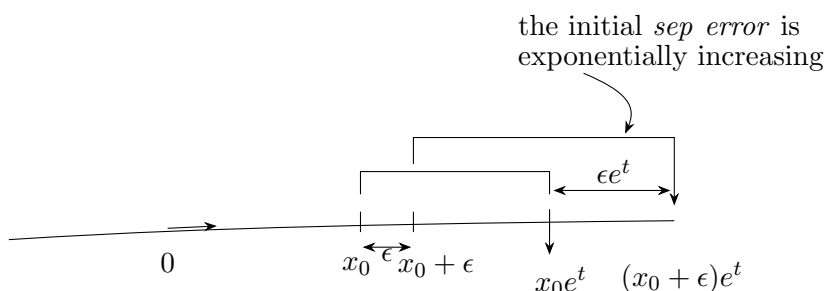
Typically, mixing occurs exponentially fast.



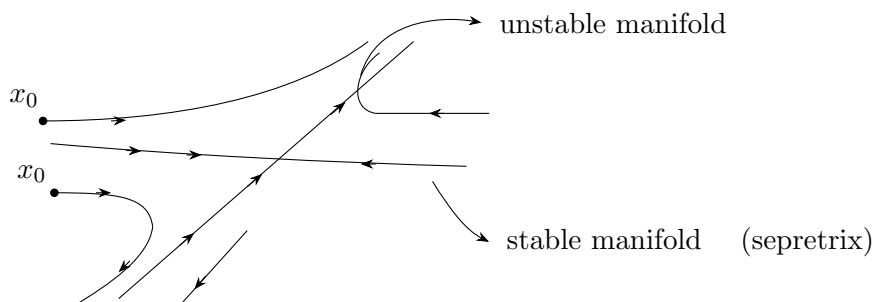
system is said to be in, exponential sensitivity to initial conditions.

example $(1 - D \text{ phase-space}) \rightarrow \text{hypothetical}$

$$\dot{x} = x \quad x(t) = x(0) e^t$$



This is just an analogy to understand but keep in mind $\dot{x} = x$ is integrable and this is not chaos.



we can see that x_0 decides the future of two sm phase space points is very different. which are in neighbourhood.

so it is like if we have sepretrix at every point that would imply chaos.

conditions of having chaos.

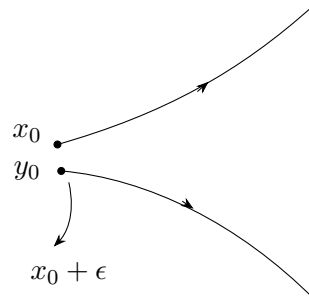
- (1) Bounded –phase space.

- (2) Exponential sensitivity to initial conditions.
- (3) A dense set of unstable periodic orbits.

Exponential sensitivity in bounded –phase space :-

If we have two different initial condⁿ x_0, y_0 with different phase trejectories. Find separation between $x(t), y(t)$.

$$|x_0 - y_0| = \epsilon$$



$$\lim_{t \rightarrow \infty} \frac{1}{t} \ln \frac{|x(t) - y(t)|}{|x(0) - y(0)|} = 0$$

as $t \rightarrow \infty$

$$|x(t) - y(t)| \leq L$$

↓
system
size
(bounded)

to avoid such situations let us take $\epsilon \rightarrow 0$

$$\lim_{\epsilon \rightarrow 0} \lim_{t \rightarrow \infty} \frac{1}{t} \ln \frac{|x(t) - y(t)|}{|x(0) - y(0)|} = \lambda(x_0) \quad 0$$

$$|x(0) - y(0)| = \epsilon$$

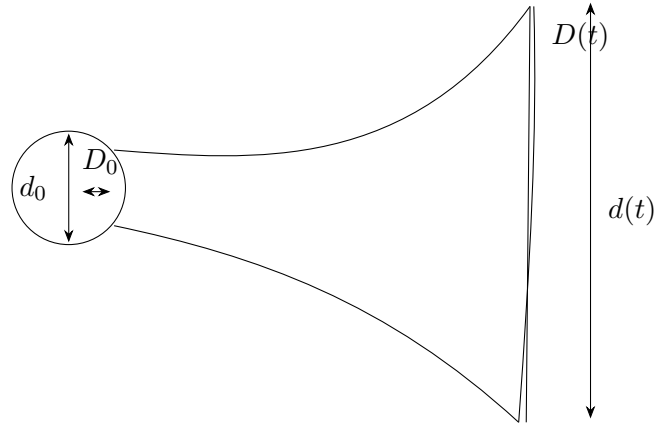
but lets invert the limits,

$$\lim_{t \rightarrow \infty} \lim_{\epsilon \rightarrow 0} \frac{1}{t} \ln \frac{|x(t) - y(t)|}{|x(0) - y(0)|} = \lambda(x_0) \quad - \textcircled{1}$$

⊥ ‘Liapunov exponent’

For n -diamentional phase space we need n -liapunov exponent, so there is a full spectrum of liapunov exponents.

eqn $\textcircled{1}$ provides maximum Liapunov exponent. as if a circle shrinks to line



(d_0 has exponential increment while D_0 has decreased with time.).

In continous time dynamics, $N = 3$ is minimum required for chaos.

For descrete time dynamics $N = 1$ will do.

$N = \text{d.o.f. Particle}$

Instead of differential eqn we call them difference eqn difference eqn are called maps.

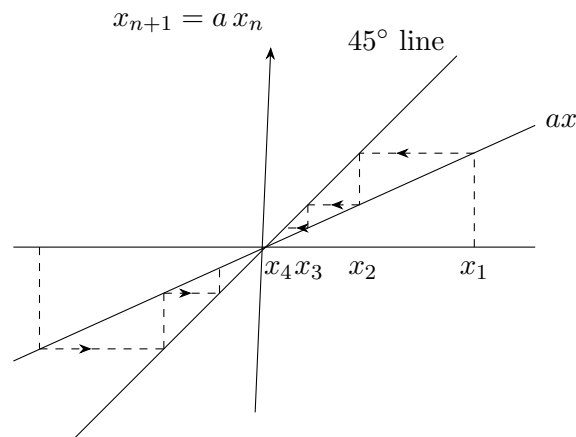
1 - D map :-

$$(\Delta x_n = x_{n+1} - x_n = a.)$$

$$x_{n+1} = f(x_n) \quad n = 0, 1, 2 \dots \text{ discrete time}$$

(i) $(a < 1)$

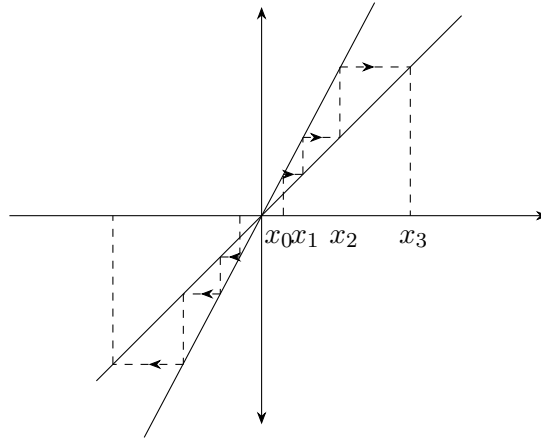
suppose $a < 1$



So, $x = 0$ is a stable fixed point

as we went to $x = 0$ from both side

(ii) $(a > 1)$

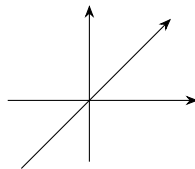


So, $x = 0$ in this case is unstable fixed point.

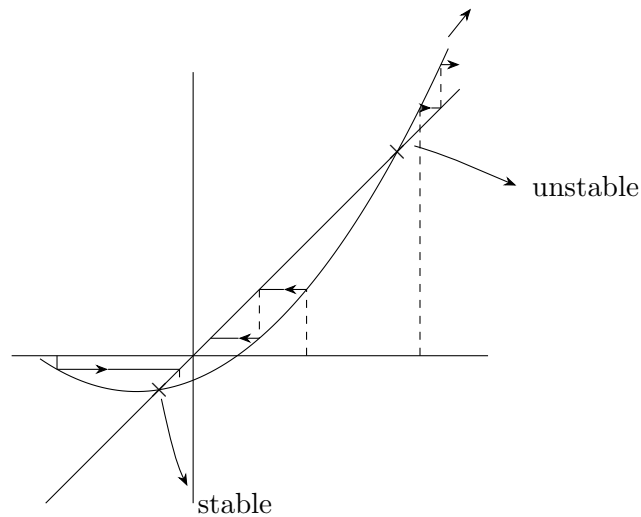
(iii) $a = 1$ map is $x = x$ (identity map)

(trivial degenerate case)

Where ever the particle is, it does not move at all.



(iv) for any arbitrary map.



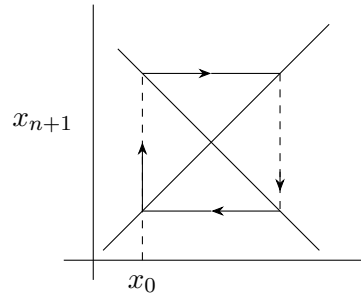
we can also see that slope at stable point is < 1 and slope at unstable fixed point is > 1 .

so, if ' x^* ' is fixed point then,

$$|f'(x^*)| > 1 \Rightarrow x^* \text{ is unstable}$$

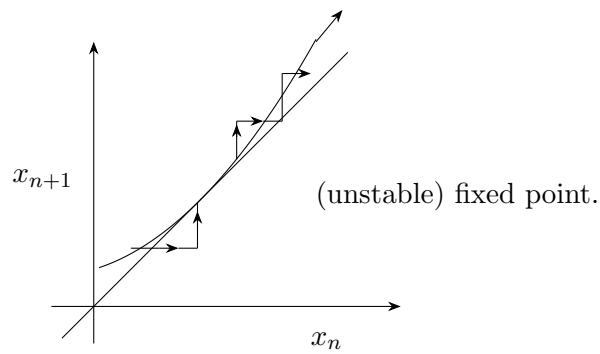
$$|f'(x^*)| < 1 \Rightarrow x^* \text{ is, stable}$$

for $f'(x^*) = -1$ *cyclic* (loop).



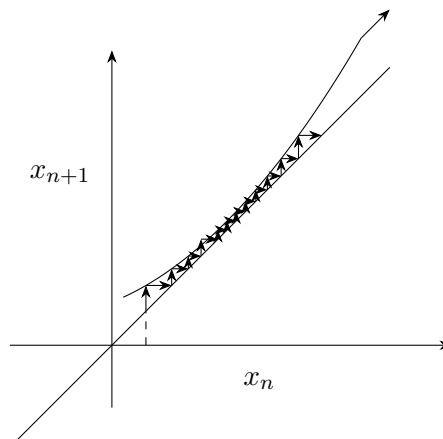
x^* = marginal fixed point (indifferent fixed point).

(v) Tangent case:-



(vi)

(fixed point is missed)



(Intermittency)

(Irregularity followed by long burst of regularity)

4 Lecture 16

Exponential divergence of phase trajectory can be characterised by “liapunov exponent”.

Note :-

we can find finite time “liapunov exponent” but it is not relevent for the purpose we have in mind.

So the reason why we need very long time liapunov exponent is following

If we start with variable x_0

$x_n = x_0$ after ‘n’ time steps.

$$x_0 \longrightarrow x_1 \longrightarrow x_2 \longrightarrow \cdots x_n$$

x_n is not computable as error is exponentially increasing

$$\downarrow \quad (x_0, x_n)$$

ex. If we write no. in binary and lets say error doubles each steps then error from 100^{th} decimal *ste* place comes to first decimal place after 100 steps.

If ‘x’ represents any physical quantity, there may be some function of ‘x’ $\rightarrow \phi(x)$ whose avg. can be written as $\langle \phi(x) \rangle$

$$\langle \phi(x) \rangle = \frac{1}{n} \sum_{j=0}^{n-1} \phi(x_j)$$

$$n \rightarrow \text{time} \quad \langle \phi(x) \rangle = \text{time average}$$

$\phi(x_0)$	$\phi(x_1)$	$\phi(x_2)$	$\phi(x_n)$
t_0	t_1	t_2	$-t_n$

since x_j is not computable, then $\langle \phi(x) \rangle$ is meaningless. but we would like to know what is long time avg

$$\langle \phi(x) \rangle = \lim_{n \rightarrow \infty} \frac{1}{n} \sum_{j=0}^{n-1} \phi(x_j) \quad - \textcircled{1}$$

$$\downarrow \quad \text{(Replacing time avg by ensamble avg.)}$$

$$\langle \langle \phi(x) \rangle \rangle = \int dx \rho(x) \phi(x) \quad - \textcircled{2}$$

If the system is ergodic; $\textcircled{1} = \textcircled{2}$.

$$\text{time avg.} = \text{ensamble avg.}$$

Note :- $\rho(x)$ should not vary with evolution and to attain this the dynamics has to seen very very long time, thats why we need long time avg also for this reason we need liapunov exponent for very long time

Non linear Map :-

simplest is $\rightarrow$ Bernoulli map/(shift).

defⁿ :-

(Bernoulli map is the map of unit interval to itself.)

x_0 lies in $[0, 1]$

If x_1 is $f(x_0)$; x_1 lies in $[0, 1]$

$$x_0 \longrightarrow x_1 \longrightarrow x_2 \longrightarrow \dots \longrightarrow x_n$$

$$x_0 \in [0, 1]$$

$$x_1 \in [0, 1] \Rightarrow x_n \in [0, 1]$$

ex :- $x_{n+1} = 2x_n \bmod 1$

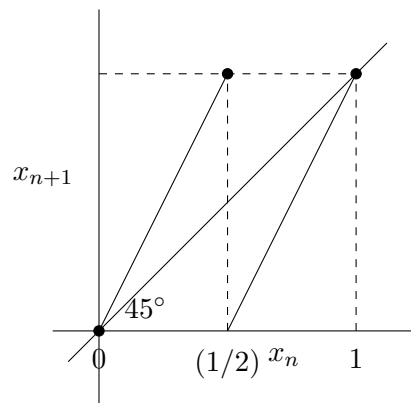
In map of $f(x)$ fixed point x^* satisfies :-

$$x^* = f(x^*)$$

fixed points are given by intersection of x_{n+1} with 45° line. which are $\{0, 1\}$

$$f'(0) = f'(1) = 2 > 1$$

so, $x^* = 0, 1$ are unstable fixed points.



This can also happen

$$f(a) = b$$

$$f(b) = a$$

first iterator of above map.

$$f^{(2)}(x) = 4x \bmod 1$$

now we have four stable points $\{0, 1/3, 2/3, 1\}$

slope at each point is 4. which is greater than 1 so fixed points are unstable

so its the non-linearity which leads to chaos.

“Any rational no is part of periodic orbit”.

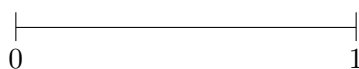
ex $x_0 = 1/3 = 0.01010101 \dots$

$$x_1 = \frac{2}{3} = 0.1010101$$

$$x_2 = \frac{1}{3} = 0.01010101 \dots = x_0 \Rightarrow \text{period '2' cycle}$$

→ (repetitive in pattern or terminating.)

Because every point is unstable, every rational no is unstable, it should act like a seperatrix which throws out no. on each side.



on line from 0 to 1 there are so many irrational no. than rational no.

and if $x_0 =$ irrational no. it wanders every where *and* by filling this interval $[0, 1]$ densely.

→ The system is ergodic.

→ The system is dissipative (How ?) → will prove later

★ Liapunov Exponent for (1-D map) :-

$$\lambda(x_0) = \lim_{n \rightarrow \infty} \lim_{\epsilon \rightarrow 0} \frac{1}{n} \log \frac{|x_n - y_n|}{|x_0 - y_0|} \quad |x_0 - y_0| = \epsilon$$

$$= \lim_{n \rightarrow \infty} \left(\lim_{\epsilon \rightarrow 0} \frac{1}{n} \ln \left| \frac{f^n(x_n) - f^n(x_0 + \epsilon)}{x_0 - (x_0 + \epsilon)} \right| \right)$$

$$f(x) = ax$$

$$f'(x) = a$$

$$a \in [0, 1]$$

$$f'(x) < 1 \Rightarrow \text{stable fixed points.} \quad (0 < a < 1)$$

also for, liapunav exponent $\lambda(x_0) = \ln a < 0$ so for stable fixed point liapunav exponent becomes negative (everything falls to one place).

Q.) what would be liapunav exponent for period 2 cycle?

$$f(a) = b$$

$$f(b) = a$$

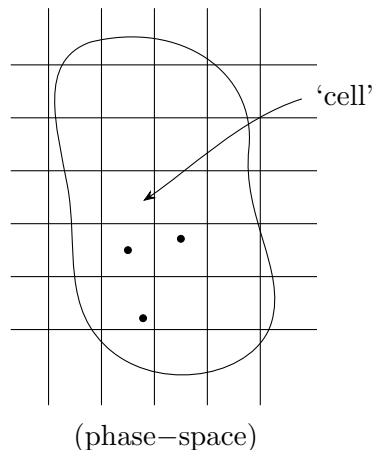
$$\text{for stability} \quad |f'(a) f'(b)| < 1$$

(slope < 1)

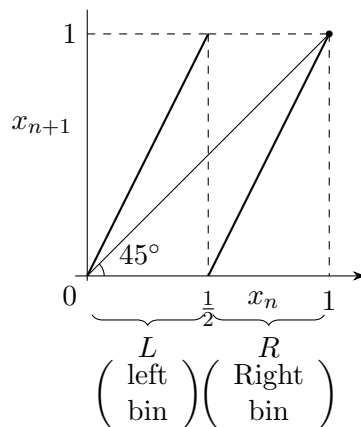
⇒ liapunov exponent will be negative for period 2 cycle.

Note :- The map is random, in the following sense, we agree that the rule is completely deterministic. for each x_0 we have well defined x_1 (with no randomness) and yet this is as random as coin toss.

suppose below is our phase space. In reality we need bins/cells to keep track of phase space points. so we do partition of phase space into cells/bins and numerically calculate how many times the perticular point lies in perticular cell/bin.



For $x_{n+1} = 2x_n \bmod 1$



we just want to know a particular point lies in left bin or right bin.

Any irrational number from 0 to 1 can be written as non-terminating in binary

$$x_i = 0.0110011111001010111000001001 \dots$$

$$x_0 = 0.a_0 a_1 a_2 \dots a_i \dots a_n$$

$$x_1 = a_0.a_1 a_2 a_3 \dots a_{i+1} \dots a_{n+1} \dots a_n$$

Above x_i is equivalent in asking wheather a_i is '0' or '1'

$$a_i = 0 \quad \text{for} \quad \longleftrightarrow L$$

$$a_i = 1 \quad \text{for} \quad \longleftrightarrow R$$

$$\text{for } x_i < \frac{1}{2} \quad a_0 = 0 \quad \longleftrightarrow L$$

$$1 > x_i > \frac{1}{2} \quad a_0 = 1 \quad \longleftrightarrow R$$

So any arbitrary string of irrational no. can be written as

$$L L R L R R R L L \dots L R L \dots L L R \dots$$

which can then also be said to be obtained from coin toss experiment.

$$\rho_2(x_2) = \int \delta(x_2 - f(x_1)) \cdot \delta(x_1 - f(x_0)) dx_1$$

Moral of the story

any given point; $y \xrightarrow{\text{iteration}} f(y)$

after 'n' time steps whats the probability density in x

$$\begin{aligned} \rho_n(x) &= \int dy \delta(x - f(y)) \rho_{n-1}(y) \\ \downarrow \lim(n \rightarrow \infty) \quad (n = \text{time steps}) \\ \left(\begin{array}{l} \text{Frobenius -} \\ \text{Perron equation.} \end{array} \right) \quad \boxed{\rho(x)} &= \int dy \delta(x - f(y)) \rho(y) \\ &\hookrightarrow \text{invariant density.} \end{aligned}$$

$\rho_n(x)$ = density of n^{th} iterate does not depends on initial condition.

—→ singular integral equation with kernel $K(x, y) = \delta(x - f(y))$

Integral eqⁿ :-

$$\phi(x) = \int dy K(x, y) \phi(y)$$

$K(x, y)$ = kernel of the integral operator.

where integral operator is $\int dy K(x, y)$ acting on $\phi(y)$

—→ Eigen value = 1

Normalization condition on $\rho(x)$

$$\int dx \rho(x) = 1$$

[?]

$$\rho(x) \geq 0, \quad x \in I$$

$$f(y) = 2y \bmod 1$$

$$\rho(x) = \int_0^1 dy \delta(x - f(y)) \rho(y) = \begin{cases} 2y & 0 \leq y < \frac{1}{2} \\ 2y - 1 & 1 \geq y > \frac{1}{2} \end{cases}$$

$$\rho(x) = \int_0^{1/2} dy \delta(x - 2y) \rho(y) + \int_{1/2}^1 dy \delta(x - 2y + 1) \rho(y)$$

$$\begin{aligned}
& \left\{ \delta(ax) = \frac{1}{|a|} \delta(x) \right. \\
& = \left(\frac{1}{2} \right) \int_0^{1/2} dy \, \delta\left(y - \frac{x}{2}\right) \rho(y) + \frac{1}{2} \int_{1/2}^1 dy \, \delta\left(y - \frac{x+1}{2}\right) \rho(y) \\
& \rho(x) = \frac{1}{2} \left[\rho\left(\frac{x}{2}\right) + \rho\left(\frac{x+1}{2}\right) \right] \quad \dots \quad \text{functional equation}
\end{aligned}$$

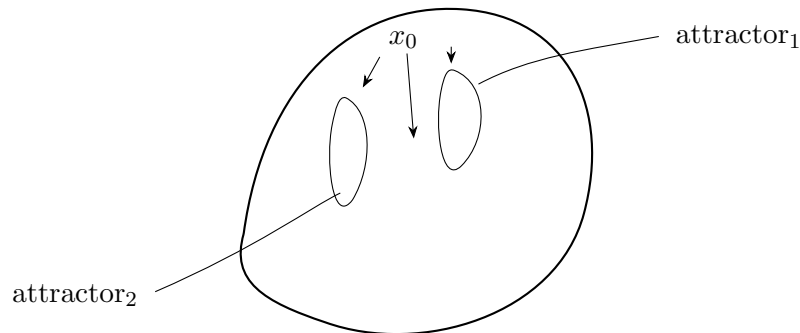
$$\rho(x) = \text{const from normalizing} \quad \int_0^1 c \, dx = 1 \quad (c = 1)$$

$$\boxed{\rho(x) = 1}$$

Therefore iterates of irrational no. uniformly and densely fill up the complete interval.

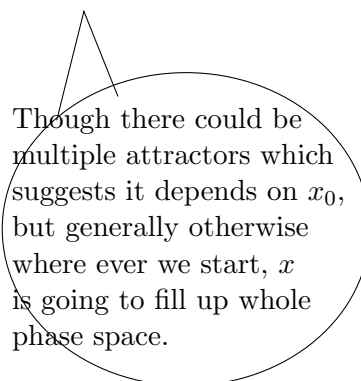
5 Lecture 17

(Q) Does Liapunov exponent depend on the initial condition or not?



If the initial condition is such that the trajectory comes to an arbitrary point in phase space.

Since the Liapunov exponent is calculated by taking time steps to infinity, and the point has explored the whole phase space, therefore in that sense the Liapunov exponent should not depend on the initial condition.



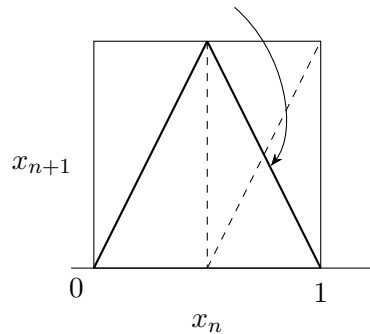
So the Liapunov exponent now becomes the property of the trajectory rather than of a point.

Baker's map :-

(measure (area, volume) is preserved, still system is chaotic), conservative system.

It is like bernoulli map where we have to cut the graph above $x_{n+1} = 1$ and put it back. We could have also folded the peice back rather than cutting to gain non-linearity.

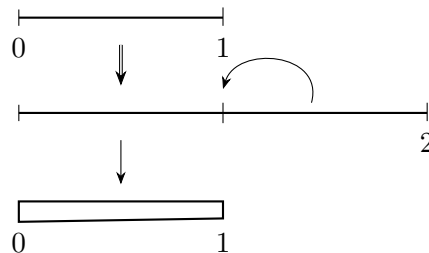
$$x_{n+1} = 2x_n \bmod 1 \quad \longrightarrow \quad (\text{fold just}) \quad (\text{analogy})$$



Tent map (non-linear, contineous)

$\lambda(x_0) = \ln 2$, all rational no are periodic unstable orbits.

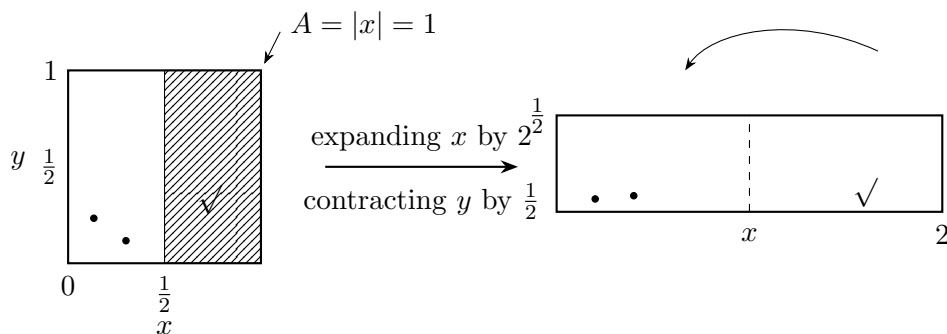
So the idea is to stretch the interval then put the second half part on top of first half map.



Backer's map takes one more diamencion (increasing dimension by one) in such a way that measure is preserved. $x_{n+1} = 2x_n \bmod 1$ [?]

$$y_{n+1} = \begin{cases} \frac{1}{2} y_n & (x_n < \frac{1}{2}) \\ \frac{1}{2} + \frac{1}{2} y_n & (x_n > \frac{1}{2}) \end{cases}$$

lets draw, x_{n+1} vs y_{n+1} ; and x_n vs y_n .

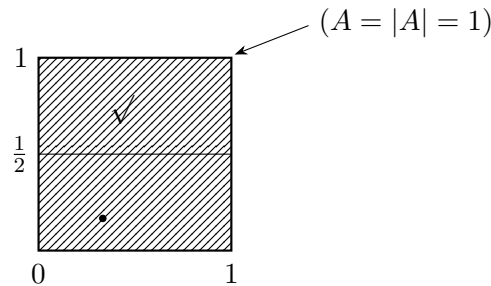


- map is invertible, still chaotic because of ($\lambda_1 = \ln 2$)
- Area is preserved ($A_i = 1, A_f = 1$)

$\Rightarrow$ Jacobian of transformation is 1

- liapunov exponents are $\ln 2, \ln \frac{1}{2}$

$$\text{total liapunov exponent} = \ln 2 + \ln \frac{1}{2} = 0$$



As long as measure is preserved sum of liapunov exponents is zero

All liapunov exponents are zero for integrable hamiltonian syst[?] since it is integrable none of $\lambda(x_0)$ should be +ve, -ve (expand[?] volume space/contracting), sum of these also will be zero, $(0 + 0 + \dots)$

For non-integrable hamilton systems liapunov exponents occur in pairs such that total liapunov exponent is zero. (-ve for each +ve).

Note :-

unlike differential dynamics, in discrete time dynam[?] we can write explicit solution to a map, but stil[?] it will be chaotic.

$$\begin{aligned} x_{n+1} &= 2x_n \bmod 1 \\ x_n &= 2^n x_0 \bmod 1 \quad \longrightarrow \text{explicit fun. of time} \\ &\quad \hookrightarrow \text{still chaotic.} \end{aligned}$$

we can see how we do not see loss of information in Backers map, if we represent $x_n, y_n, x_{n+1}, y_{n+1}$ in binary

$$\begin{aligned} x_n &= 0.a_0 a_1 a_2 a_3 \dots \\ y_n &= 0.b_0 b_1 b_2 b_3 \dots \\ &\downarrow \\ x_{n+1} &= 0.a_1 a_2 a_3 \dots \\ y_{n+1} &= 0.(a_0 b_0) b_1 b_2 b_3 \dots \end{aligned}$$

so we do not loss 'a₀'.

Chaotic but (measure preserving and invertible.)

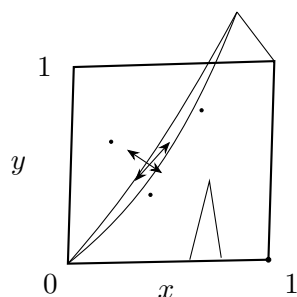
(wow!)

Note :- The general rule to produce chaos in 1-d map is that it has to be non-invertible, for 2 and higher diamen-ncions invertible maps can also produce chaos.

Arnold's cat map :-

$$\begin{aligned}x_{n+1} &= (x_n + y_n) \bmod 1 \\y_{n+1} &= (x_n + 2y_n) \bmod 1 \\ \begin{pmatrix} x_{n+1} \\ y_{n+1} \end{pmatrix} &= \begin{pmatrix} 1 & 1 \\ 1 & 2 \end{pmatrix} \begin{pmatrix} x_n \\ y_n \end{pmatrix}\end{aligned}$$

determinant of Transformation matrix = 1 (orientation preserving)



The reason it is named as arnold's cat is because he put face of cat in this graph and after few iteration we can't recognize the face of cat.

→ every point is having stretching direction and contracting direction.

***. Gauss (continued fraction) map :-**

$$x_0 = \frac{1}{a_0 + \frac{1}{a_1 + \frac{1}{a_2 + \frac{1}{a_3 + \frac{1}{a_4 + \dots}}}}} \xrightarrow[\text{writing}]{\text{eas of}} \frac{1}{a_0} \oplus \frac{1}{a_1} + \frac{1}{a_2} + \frac{1}{a_3} \dots$$

↔ notice '+' sign is written down

If continued fraction terminates x_0 = rational else x_0 is irrational number

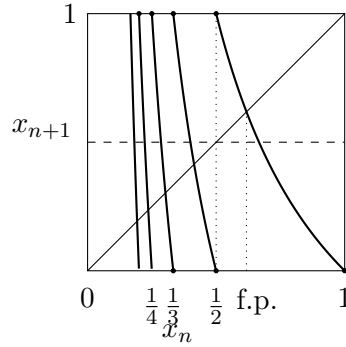
$$x_1 = \frac{1}{a_1 + \frac{1}{a_2 + \frac{1}{a_3 + \frac{1}{a_4 + \dots}}}}$$

So to get x_1 ; take reciprocal of x_0 , throw away integer part ' a_0 '. So,

$$x_{n+1} = \frac{1}{x_n} - \left[\frac{1}{x_n} \right]$$

↔ integer part of $(1/x_n)$

$$\text{for, } \begin{pmatrix} \frac{1}{2} < x_n < 1 \\ 1 < \frac{1}{x_n} < 2 \end{pmatrix}$$



for any given x_{n+1} there lies infinite pre images.

|slope at each point| > 1

→ unstable at each point.

Q.1) lets take any arbitrary no from 0 to 1. take its reciprocal throw away integer, and repeat the process for long time, what is the probability that final number is less than α . $(0, \alpha)$
 (Ans: $\frac{\ln(1+\alpha)}{\ln 2}$)

Before we calculate let's find fixed point of gauss map.

$$\text{f.p.} \Rightarrow x = \frac{1}{x} - 1$$

$$x^2 + x - 1 = 0 \quad x = \frac{-1 \pm \sqrt{1+4}}{2} \\ = \frac{-1 \pm \sqrt{5}}{2}$$

$$\text{So f.p.} = \frac{\sqrt{5}-1}{2} \quad \text{as} \quad 0 < x < 1$$

$$\rightarrow x + 1 = \frac{1}{x} \\ \rightarrow x = \frac{1}{1+x}$$

continued fraction can be written as

$$x = \frac{1}{1 + \frac{1}{1 + \frac{1}{1 + \dots}}}$$

For f.point₂ , $x = \frac{1}{x} - 2$

$$\Rightarrow x^2 + 2x - 1 = 0 \\ x = \frac{-2 \pm \sqrt{4+4}}{2} = \frac{-2 \pm 2\sqrt{2}}{2} \Rightarrow -1 \pm \sqrt{2} \\ \left(x = \sqrt{2} - 1 \right)$$

$$\begin{aligned}
\text{continued fraction} \Rightarrow \quad & \frac{1}{x} = x + 2 \\
\Rightarrow \quad & x = \frac{1}{x + 2} \\
\Rightarrow \quad & x = \frac{1}{2 + \frac{1}{2 + \frac{1}{2 + \dots}}}
\end{aligned}$$

Q.) what short of this no is?

$$x = \frac{1}{1 + \frac{1}{2 + \frac{1}{1 + \frac{1}{2 + \dots}}}}$$

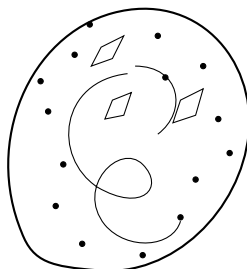
Ans. It would be fixed point of first iterate of map or period 2 cycle.

6 Lecture 20: (classical statistical mechanics)

- statistical mechanics helps in understanding behaviour of systems which involves large no of d.o.f (which are interacting non-linearly with one another).
- It is science of averages. Also it is study at largest/longest length scales and time scales.
- statistical mechanics is partitioned into two parts
 - Equilibrium statistical mechanics.
 - Non-Eqm statistical mechanics.
- Eq^m statistical mechanics arose from question that :- Are there any steady states (eqⁿ states) which the system will finally settle down? If there is How does it behave and so on?
- Thermodynamics is the study of systems at their longest length scales and time-scales.
 - so All fluctuations which happens at very small length scales & time scales are neglected and we look at things at very long time scales and very long length scales.

let us start with hamiltonian of a system. system contains many particles and is isolated from the rest of the universe.

Isolated system (exchange of matter & energy are prohibited).



$$H(q, p)$$

If rules are time-translation invariant then we know that Hamiltonian is constant of motion.
 $\left(\frac{dH}{dt} = 0\right)$

$$H(q, p) = E \quad (\text{total energy of system}).$$

{with uncertainty δE }

phase space becomes $6n$ -dimensional for n -particles. So we need to know $3n$ – constants of motion which are in involution with each other. $\{c_i, c_j\} = 0 \quad i \neq j$.

constants of motion will be

1) H	1
2) total linear momentum (P)	3
3) Total angular momentum (L)	3
4) C.o.M position. ($\vec{R}$)	3
Total	<u>10 COM</u>

To specify state of system we need a point in phase space $(q_1, q_2, \dots, q_n, p_1, p_2, \dots, p_n)$. And that point will move on some $6n$ -dimensional hypersurface.

- Due to lack of sufficient nos of c.o.m which are in involution with each other, we know that differential eqⁿ of motion is not integrable and the system of such kind is going to show chaotic behaviour with some positive liapunov exponents. Its likely to wander around completely in strange fashion. such that system is ergodic on energy surface.
- It does not have to be strictly ergodic, there can be great deal of mixing (exponential separation of volume element).

Any way exponential separation $\Rightarrow$ mixing (implies.)

$\Downarrow$

ergodicity

Ergodicity :–

- The volume element of phase space will come closer to every point of phase space in given sufficient time. {defⁿ of ergodicity}.
- The whole phase space is not accessible to the system because we specified its total energy.
- we become ignorant in knowing what happens to system at time ' $t_0 + \Delta t$ '. and assume that every point/volume element of phase space is as equally likely as other points of phase space. We can term this principle as principle of maximum ignorance.

Fundamental Assumption/postulate of eqⁿ statistical mechanics.

- “In a state of thermal eq^m, all accessible micro-states of the system are equally probable”.

“This single statement is enough to derive all equations of stat. mech, whereas thermal eqⁿ turns out to be a special case of it”.

Thermal equilibrium :—

Two systems are said to be in thermal equilibrium if there is no heat flows between them when they are connected to each other.

A system is said to be in thermodynamic eq^m if it is in mechanical, physical and thermal equilibrium.

- state of thermal eq^m also means that long time average of macroscopic variables reach to a steady state and do not change with time. such macroscopic quantities are Energy (internal), Total pressure[?]

“All physical macroscopic quantities are time independent in system of thermal eq^m state.”

Note :—

- Microscopic variables like velocity of a particle in certain d.o.f is not time independent, whereas the average velocity of all particles is time independent which is zero. i.e. system is not moving anywhere.
- The average values can only be independent of time if the probability distribution (over which average is defined) is itself independent of time.

micro state :—

specifying the state of every constituent of the system. The set $(q_1 \cdots q_n, p_1 \cdots p_n)$ is microstate of the system.

or we can say that position in phase space is microstate where as the total (avg) momentum of all particles is macrostate.

By accessible microstate we restrict the velocity a particle can have, i.e.

$$H(q, p) = E_{\text{total}}$$

so a particle of such isolated system can not have $K.E > E$. So some micro states are not accessible anymore.

“All those micro states which are compatible with all restrictions applied on system are termed as accessible micro-state.”

“microstate tells what each and every particle is doing and macrostate is giving some gross (overall) information.”

If microstate energy is the only parameter of microstate — (In quantum stat mech)

“If a isolated system has 10J of total energy then we can have a single particle which has energy close to 10J and rest. of all are close to zero energy. This (microstate) is equally probable than other state; followed from postulate of eq^m stat. mech.”

In phase-space probability density obeys

$$\frac{\partial \rho}{\partial t} = \{H, \rho\}$$

$\rho = \text{const. of motion}$

$$\frac{d\rho}{dt} = \{\rho, H\} + \frac{\partial \rho}{\partial t} = 0$$

$$\frac{\partial \rho_{eq}}{\partial t} = \{H, \rho_{eq}\}$$

if probability distribution is independent of time

$$\frac{\partial \rho_{eq}}{\partial t} = 0 = \{H, \rho_{eq}\} \Rightarrow \text{implies } \underline{\rho_{eq} = f(H)}.$$

remember ρ_{eq} is not physical observable.

$$\rho_{eq} = \rho_{eq}(H) = \delta(H(q, p) - E)$$

since such isolated system uses microstates, such ensemble is termed as microcanonical ensemble. A stat. ensembles with $\rho_{eq} = \delta(H(q, p) - E)$ in phase space with which corresponds to isolated system in thermal eqⁿ is called microcanonical ensemble

> If we assume we can not have a point in phase space but a volume shell of finite resolution then only we can define probability of occupying certain state (volume element in phase space (shelle)).

$$\Delta q \Delta p \geq \frac{\hbar}{2}$$

for 1 particle volume element

$$(\Delta q_1 \Delta p_1)(\Delta q_2 \Delta p_2)(\Delta q_3 \Delta p_3)$$

$$\sim \hbar^3 \text{ or } h^3$$

for N particle the volume element would be

$$\sim h^{3N}.$$

If volume of accessible phase space is ' M ',

$$\# \text{ of micro states} = \frac{M}{h^{3N}} = \Omega(E) \text{ (finite \& large)}$$

let probability of any microstate = P $\sum_i P_i = 1$

$$P \cdot (\text{no. of microstates}) = 1$$

$$\Rightarrow \left\{ P = \frac{1}{\Omega} = \frac{h^{3N}}{M} \right.$$

Toy Model :-

set of N distinguishable (identifiable) coins. where each coin has $\{H, T\}$ two states.

$$\{2^N \text{ microstates}\} \quad \underbrace{\text{htt} \quad - \quad \text{hhh} \quad - \text{tth} \quad - \quad \text{hhh}}_{\text{'N' letters}} \longrightarrow \text{microstate}$$

(still a state
of system as
a whole)
but describes
what each object is
doing

Let,

Total no. of heads = H

of Tails = T

$$H + T = N$$

Let,

$$H - T = M$$

(N, M) or (N, H) or (H, T) are macrostate of the system.

For given N ,

$$\# \text{ of macrostates} = N + 1.$$

ex for $N = 3$ let (N, H) represents macrostate

we have: $(3, 0), (3, 1), (3, 2), (3, 3)$ 4 macrostates.

no. of microstates are much larger than no. of macrostates.

$$2^N > N + 1$$

$$N = d.o.f.$$

> probability of any possible/specific microstate would be $\left(\frac{1}{2^N}\right)$.

> Probability of attaining a given macrostate :-

we have $0 \leq H \leq N$

$$P(H) = {}^N C_H \cdot \left(\frac{1}{2}\right)^H \left(\frac{1}{2}\right)^{(N-H)}$$

Suppose coin is biased then

$$P(h) = p$$

$$P(t) = 1 - p = q$$

then,

$$P(H) = {}^N C_H p^H q^{N-H}$$

$$\sum_{H=0}^N P(H) = (p + q)^N = 1^N = 1 \quad \text{binomial expansion.}$$

Generating function for the binomial distribution :-

$$\begin{aligned} f(x) &= \sum_{H=0}^N P(H) x^H \\ &= (px + q)^N = \sum_{H=0}^N ({}^N C_H p^H q^{N-H}) x^H \end{aligned}$$

$P(H)$ = coefficient of x^H in expanded generating function

Property of G.F :-

$$f(1) = 1 ; (p + q)^N = 1$$

$$\langle H \rangle = ? \quad (\text{avg. no. of Heads})$$

$$\langle H \rangle = \frac{\sum_{H=0}^N P(H) H}{\sum_{H=0}^N P(H)} = \frac{\sum_{H=0}^N P(H) H}{1}$$

to find $\langle H \rangle = \sum_{H=0}^N P(H) H$ we differentiate G.F

$$f'(x) = \sum_{H=0}^N P(H) \cdot H x^{H-1}$$

$$f'(1) = \sum_{H=0}^N P(H) \cdot H = (N(px + q)^{N-1} \cdot p)_{x=1}$$

so,

$$\langle H \rangle = \sum_{H=0}^N P(H) H = N(p + q)^{N-1} \cdot p = Np$$

$$\boxed{\langle H \rangle = Np}$$

$$\langle H^2 \rangle = ?$$

find

$$f''(1) = (Np^2(N-1)(px + q)^{N-2})_{x=1} = p^2 N(N-1) \quad (1)$$

$$f''(x) \Big|_{x=1} = \left(\sum_{H=0}^N P(H) H(H-1) x^{H-2} \right)_{x=1} = \langle H(H-1) \rangle$$

$$= \langle H^2 - H \rangle$$

$$= \langle H^2 \rangle - \langle H \rangle \quad (2)$$

Equating (1) and (2)

$$p^2 N(N-1) = \langle H^2 \rangle - \langle H \rangle$$

$$\Rightarrow \langle H^2 \rangle = p^2 N^2 - p^2 N + \langle H \rangle$$

$$\Rightarrow \langle H^2 \rangle = p^2 N^2 - p^2 N + Np$$

Also, Now we can find variance,

variance = mean square – square of mean

$$\text{var}(H) = \langle H^2 \rangle - \langle H \rangle^2 \quad (= \langle (H - \langle H \rangle)^2 \rangle)$$

$$= N^2 p^2 - p^2 N + Np - N^2 p^2$$

$$\text{var}(H) = Np(1-p)$$

$$\sigma^2 = \text{var}(H) = Npq \quad , \text{ as, } (q = 1-p)$$

Therefore $\Delta H = \text{std. deviation} = \text{uncertainty} = \sqrt{Npq}$

$$\boxed{\Delta H = \sigma = \sqrt{Npq}}$$

Therefore relative fluctuation will be :-

$$\text{fluctuation about mean} = \frac{\Delta H}{\langle H \rangle} = \frac{\sqrt{Npq}}{Np} = \sqrt{\frac{q}{Np}}$$

$$\text{Relative scatter} = \frac{1}{\sqrt{N}} \sqrt{\frac{q}{p}}$$

(dispersion about mean)

> It says relative fluctuation becomes smaller as no. of coin toss increases / no. of coins increases.

In thermodynamics we deal with $\sim 10^{23}$ d.o.f so relative fluctuations becomes very small. This is the main reason “why thermodynamics works”.

“In statistical mechanics generating function is called as partition function (laplace transformation / green fn.)”

Note :-

out of 100 coin toss $P(50H)$ or $P(50T)$ is most probable. Because of ${}^{100}C_{50}$ is biggest for all H in $({}^{100}C_H)$.

each microstate is equally probable, but each macrostate is made up of many microstates. so attaining 1H and 99T is having lesser no of micro-states then that of 50H, 50T So our example that any one molecule can have nearly all total energy and

Rest of all are nearly at rest; Indeed such micro-states are equally probable but the macrostate has contribution from very less no. of microstates so such macrostate is less likely to exist then other possible macrostates.

7 Lecture 21

N coins, H heads, T tails

$$P(H) = {}^N C_H p^H q^{N-H}$$

p = prob of heads

q = prob of tails

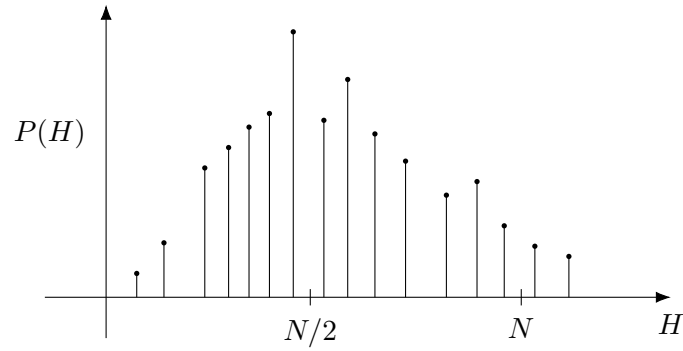
$$f(x) = (px + q)^N$$

$$\langle H \rangle = Np$$

$$\Delta H = \sqrt{Npq}$$

$$\begin{aligned} f(x) &= \sum_{n=0}^N P(n) x^n \\ &= \sum_{n=0}^N {}^N C_n p^n q^{N-n} x^n \\ &= (px + q)^N \end{aligned}$$

What does this distribution looks like.



$$P(H) = \frac{N!}{H!(N-H)!} p^H q^{N-H}$$

we would like to find out what does ${}^N C_H$ do at large no. of 'N'

What is $N!$ (for large N) :-

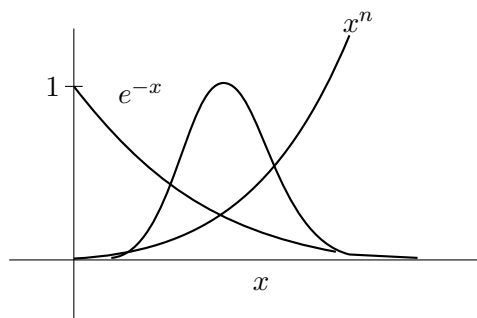
Stirling formula :-

$$\begin{aligned} n! &= n(n-1)(n-2) \dots 3 \cdot 2 \cdot 1 \\ &= n^n \left(1 - \frac{1}{n}\right) \left(1 - \frac{2}{n}\right) \dots \\ &= n^n e^{-n} \sqrt{2\pi n} \cdot \left\{1 + \frac{1}{12n} + \dots\right\} \\ 1! &= 1^1 e^{-1} \sqrt{2\pi} \quad \text{so} \quad \text{is } e \simeq \sqrt{2\pi} ? \end{aligned}$$

defⁿ

$n!$ is defined as

$$\int_0^\infty dx e^{-x} x^n = \Gamma(n+1) \quad (n = 1, 2, 3, \dots)$$



$$\begin{aligned} g(x) &= e^{-x} x^n = e^{-x} e^{\ln x^n} = e^{-x+n \log_e x} = e^{-(x-n \ln x)} \\ g'(x) &= e^{-(x-n \ln x)} \cdot (-1) \left(1 - \frac{n}{x}\right) = 0 \Rightarrow \boxed{g'(n) = 0} \end{aligned}$$

let, $f(x) = x - n \ln x$

$$f(n) = n - n \ln n$$

$$\begin{aligned}
f'(x) &= 1 - \frac{n}{x} \\
f''(x) &= 0 + \frac{n}{x^2} \\
f(x) &= f(n) + f'(n)(x-n) + \frac{f''(n)(x-n)^2}{2!} + \dots \\
f(x) &= (n - n \ln n) + 0 + \frac{(x-n)^2}{2n} + \dots
\end{aligned}$$

$$\begin{aligned}
n! &= \int_0^\infty e^{-x} x^n dx = \int_0^\infty e^{-(x-n \ln x)} \\
\Rightarrow n! &= \int_0^\infty e^{-\left((n-n \ln n) + \frac{(x-n)^2}{2n} + \dots\right)} dx \\
&= \int_0^\infty (e^{-n} \cdot n^n) \cdot e^{-\frac{(x-n)^2}{2n}} \{1 + \dots\} dx
\end{aligned}$$

for very large 'n'

$$\begin{aligned}
n! &= e^{-n} n^n \int_{-\infty}^\infty dx e^{-\frac{(x-n)^2}{2n}} dx \\
&= e^{-n} n^n \sqrt{2\pi n} \cdot \left\{1 + \text{Order}\left(\frac{1}{n}\right)\right\}
\end{aligned}$$

$$\int_{-\infty}^\infty e^{-\frac{t^2}{a}} dt = \sqrt{\pi a}$$

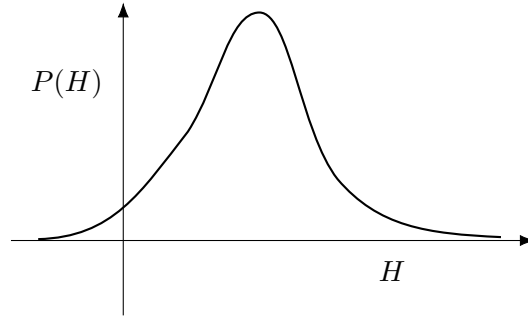
Gaussian integral

$$I = \int_{-\infty}^\infty e^{-ax^2} dx = \sqrt{\pi/a}$$

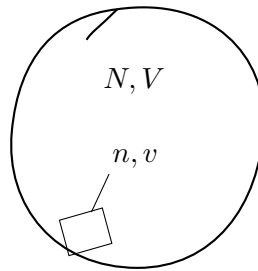
also

$$\begin{aligned}
I &= \int_{-\infty}^\infty e^{-ay^2} dy \\
I^2 &= \int_{-\infty}^\infty \int_{-\infty}^\infty e^{-a(x^2+y^2)} dx dy \quad ; \quad \begin{cases} x^2 + y^2 = r^2 \\ x = r \cos \theta \\ y = r \sin \theta \\ dA = (r d\theta) dr \end{cases} \\
&= \int \int e^{-ar^2} r dr d\theta \\
&= \int_{-\infty}^\infty e^{-ar^2} r dr \int_0^\pi d\theta = \left(\sqrt{\pi/a}\right)^2 \\
&\quad \left(I = \sqrt{\pi/a}\right)
\end{aligned}$$

so for large N



Take ideal gas contained in volume V .



N = total no. of particles.

What is probability that v contains n particles?

$P(n)$ = Probability that ' v ' contains n particle

Probability that a given particle is in volume ' v ' = $\frac{v}{V}$

assume that all the particles are moving independent of each other, therefore the probability that there are ' n ' of them inside ' v ' is

$$= {}^N C_n \left(\frac{v}{V} \right)^n \left(1 - \frac{v}{V} \right)^{N-n}$$

Here ' n ' is random variable whose no. can vary from $(0 \leq n \leq N)$.

$$\text{here } p = \frac{v}{V}, \quad q = 1 - \frac{v}{V}$$

this is the binomial distribution.

lets put $\rho = \frac{N}{V}$

$$\Rightarrow \frac{1}{V} = \frac{\rho}{N}$$

so,

$$P(n) = {}^N C_n \left(\frac{v\rho}{N} \right)^n \left(1 - \frac{v\rho}{N} \right)^{N-n} \quad \left(\begin{array}{l} \text{binomial} \\ \text{distribution} \end{array} \right)$$

What happens to $P(n)$, when $N \rightarrow \infty$ and $V \rightarrow \infty$ but keeping ' ρ ' fixed.

$$\begin{aligned} P(n) &\xrightarrow{N \rightarrow \infty} \frac{N!}{n! (N-n)!} \left(\frac{v\rho}{N} \right)^n \left(1 - \frac{v\rho}{N} \right)^{N-n} \\ &= \frac{N^N e^{-N} \sqrt{2\pi N}}{(n^n e^{-n} \sqrt{2\pi n}) (N-n)^{N-n} e^{-(N-n)} \sqrt{2\pi (N-n)}} \left(\frac{\rho v}{N} \right)^n \left(1 - \frac{\rho v}{N} \right)^{N-n} \end{aligned}$$

$$\left(\begin{array}{l} \text{we can't} \\ \text{use } n! = e^{-n} n^n \sqrt{2\pi n} \\ \text{as 'n' can take 0 too } \infty \end{array} \right) \quad (n!)$$

$$\boxed{\rho v = \bar{n}}$$

$$P(n) \longrightarrow = \frac{e^{-\bar{n}} (\bar{n})^n}{(n!)} \quad \text{where } n = 0, 1, 2, 3 \dots \infty$$

and $\bar{n} = \rho v$ = avg. no of particles in sub volume 'v'.

$$\left(\bar{n} = Np = N \frac{v}{V} \right) = (\bar{n} = \rho v)$$

we have assumed that particles behave like newtonian particle mech. and is distinguishable from one another.

In $\left(P(n) = \frac{e^{-\rho v} (\rho v)^n}{n!} \right)$ formula, (N, V) don't appear anymore, as $(N, V \rightarrow \infty)$ Such that their ratio $\rho = \frac{N}{V}$ is finite (this is called thermodynamic limit of system when $N \rightarrow \infty, V \rightarrow \infty$ such that $\rho = \frac{N}{V}$ is finite). Statistical mech. Reduce to thermodynamics in thermodynamic limit.

Poisson Distribution :-

for random var. $n = 0, 1, 2 \dots \infty$

$$P(n) = \frac{e^{-\lambda} \lambda^n}{n!} \quad (\lambda = \langle n \rangle)$$

λ is avg value of random variable

Generating function

$$f(x) = \sum_{n=0}^{\infty} P(n) x^n = e^{\lambda(x-1)} \quad (\text{How ?})$$

$$f(1) = 1 \quad \text{for} \quad \sum_{n=0}^{\infty} P(n) = 1$$

$$\text{variance} = \langle (n - \langle n \rangle)^2 \rangle = \lambda = \langle n \rangle = \langle n^2 \rangle - \langle n \rangle^2$$

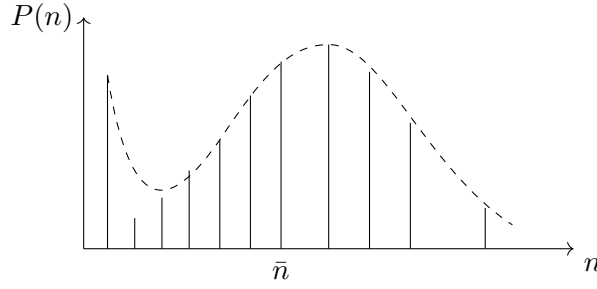
In poisson distribution variance = mean

$$\text{Relative fluctuation} = \frac{\sigma}{\langle n \rangle} = \frac{\Delta n}{\langle n \rangle} = \frac{\sqrt{\langle n \rangle}}{\langle n \rangle} = \frac{1}{\sqrt{\langle n \rangle}}$$

for poisson's distribution, $\langle n \rangle = \lambda$, $(\Delta n)^2 = \lambda$. & all higher moments = λ ., so its a single parameter distribution;

In gaussian distribution except for $\langle n \rangle$ & $(\Delta n)^2$ all higher moments vanish.

$\bar{n}$ = avg. no of particles in subvolume 'v'.



if $\bar{n}$ itself become very large then deviation from mean ($n - \bar{n}$) takes gaussian curve. and call that a continuous variable if $\bar{n}$ is very large.

from

$$P(n) = \frac{e^{-\bar{n}} (\bar{n})^n}{n!}$$

using sterling formula again, and replacing variable to

$$x = n - \bar{n}$$

$$P(x) \propto \frac{e^{-\frac{x^2}{2\sigma^2}}}{\sqrt{[?]}} \quad (\text{Gaussian distribution})$$

“ The whole point of above discussion is that if we start with a binomial distribution of burnoulli trials and then from that we take no. of particles / trials to be very large such that probability of succes in trial is vanishingly small. Then Binomial distribution goes over into poisson distribution. Then if mean value of poisson distribution is very large compared to unity, The deviation from the mean ($n - \bar{n}$) is approximatly a contineous variable that has a Gaussian shape”.

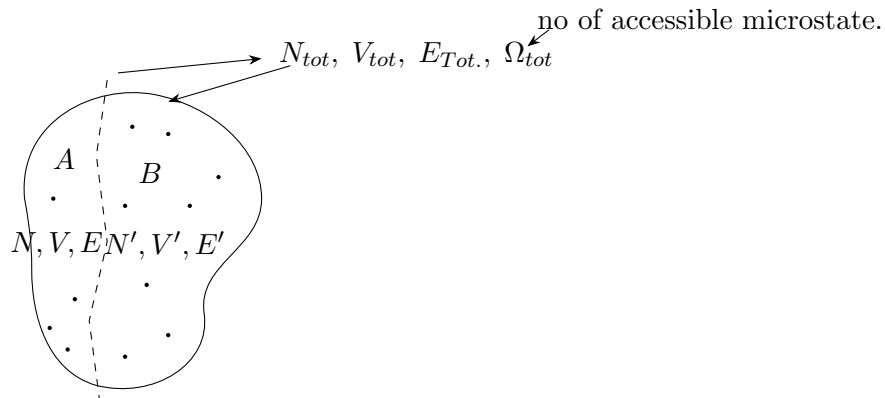
So A probability distribution can shifts over to another probability distribution.

Generating function :- $f(x) = \sum_{n=0}^{\infty} P(n)x^n = P(0)x^0 + P(1)x^1 + P(2)x^2 + \dots P(n)x^n$

$$f(x) = \sum_{n=0}^{\infty} P(n)x^n = \sum_{n=0}^{\infty} \frac{e^{-\lambda} \lambda^n}{n!} x^n \Rightarrow e^{-\lambda} \sum \frac{(\lambda x)^n}{n!} = e^{-\lambda} e^{\lambda x}$$

$$\boxed{f(x) = e^{\lambda(x-1)}}$$

lets go back to our problem. of isolated system in thermal equilibrium.



Imagine the system is made of two sub-systems, in thermal eq^m with each other, such that

$$\begin{aligned}N + N' &= N_{Tot} \\V + V' &= V_{Tot} \\E + E' &\simeq E_{tot}\end{aligned}$$

(Isolated, in thermal eq^m)

not necessarily

$(E + E' \neq E_{Tot}$ always if pot. energy plays role)

$E + E' \simeq E_{Tot}$ to a very good approximation.

“as no. of d.o.f which are interacting near partition is $(\frac{1}{10^8})^{th}$ of the total d.o.f”. (assuming short range forces”).

Then at any instant of time, what is the probability that Energy of A is E .; $P(E)$

$P(E)$ = no. of microstate of this entire system such that A has energy E . ($\Omega(E)$) (also (no. of microstates such that B has energy E' .)

$$= \frac{\Omega(E) \Omega'(E')}{\Omega_{tot}(E_{tot})} \quad (\text{normalization})$$

$\Omega(E)$ & $\Omega'(E')$ can have totally different kind of functions.

As A can be a large container of oil and B can be our atmosphere. so A, B have different d.o.f.

8 Lecture 22

Note that Energy of A and B are not fixed. Because there are fluctuations.

E, V, N, N', V', E' . are variable quantities.

what we were trying is to find, is

“Given the postulate of equal apriori probabilities of all the accessible microstates of the total system, what can we say about probability distribution of (for instance say) Energy in system A .”

$P(E) =$

$$\frac{\# \text{ of microstates of tot. system such that } A \text{ has energy } E. (\& B \text{ has } E'.)}{\text{Total no. of microstates of total system such that T.E} = E_{tot}.}$$

If A & B are assumed as independent system (they actually are not) then $\Omega_{tot}(E) = \Omega(E) \cdot \Omega'(E')$

$$= \frac{\Omega(E) \cdot \Omega'(E')}{\Omega_{tot}(E_{tot})}.$$

$P(A \& B) = P(A) \cdot P(B)$ only iff A, B are independent events

In above discussion we have assumed that interection Energy is fairly negligible.

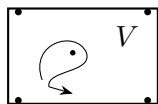
$\Omega(E)$ is actually a fun. of V, N also

Remember always

$$\begin{aligned}\Omega(E) &= \Omega(E, V, N) \\ \Omega'(E') &= \Omega'(E', V', N')\end{aligned}$$

lets see how big these no. are :

Suppose we take a single particle & put that into volume V and the particle is moving around with some energy $\leq E$. lets calculate how many microstates it has.



No. of microstates is the no. of shells in phase-space

$$\text{true in any dimensions} \leftarrow \left(E = \frac{p^2}{2m} \right) \quad |p| \leq \sqrt{2mE}$$

let no. of such microstates be $\phi(E) = \frac{\text{phase space volume}}{\text{min volume}(h^3)}$

$$\phi(E) = \frac{1}{h^3} \int_V d^3q \int_{|p| \leq \sqrt{2mE}} d^3p \quad \rightarrow \text{phase space volume}$$

$d^3p = 4\pi p^2 dp$ volume element

$$= \frac{V}{h^3} \int_0^{\sqrt{2mE}} 4\pi p^2 dp \propto \frac{V}{h^3} E^{3/2} \quad \rightarrow \text{no of dimensions}$$

$(p \propto E^{1/2})$.

$$\begin{aligned}&= \frac{V}{h^3} \frac{4\pi}{3} (p^3) \Big|_0^{\sqrt{2mE}} \\ &= \frac{V}{h^3} \frac{4\pi}{3} \cdot E^{3/2} 2m^{3/2}\end{aligned}$$

relation b/w E & p for a free particle is

$$E^2 = c^2 p^2 + m^2 c^4$$

m = rest mass always

mass does not increase with speed, its just $p = \gamma m v$ instead of mv . $\left(\gamma = \frac{1}{\sqrt{1-v^2/c^2}} \right)$

$$E = (c^2 p^2 + m^2 c^4)^{1/2} = c (p^2 + m^2 c^2)^{1/2}$$

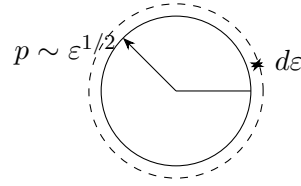
If $p \ll mc$ { mc = compton momentum

In 'n' dimentions, $(n-1)$ angles runs from $(0 - \pi$ or $0 - 2\pi)$.

In n -spacial dimen.

$$\begin{aligned}\phi(\varepsilon) &\propto \int_0^{\sqrt{2mE}} p^{n-1} dp \propto \varepsilon^{n/2} \\ E &= mc^2 \left(1 + \frac{p^2}{m^2 c^2} \right)^{1/2} \\ &= mc^2 \left(1 + \frac{p^2}{2m^2 c^2} - \frac{1}{8} \frac{p^4}{m^4 c^4} + \dots \right) \\ &= mc^2 + \frac{p^2}{2m} + \dots = \text{origin of } \frac{p^2}{2m}\end{aligned}$$

No of microstates with energy in $(\varepsilon, \varepsilon + d\varepsilon) \sim \varepsilon^{1/2} d\varepsilon = d\Phi(\varepsilon)$ (given energy shell) $p = \sqrt{2m\varepsilon}$



$$\begin{aligned}
 dV &= 4\pi p^2 dp \\
 &= 4\pi(2m)\varepsilon \frac{(2m)}{2\sqrt{2m\varepsilon}} \\
 dv &\simeq 8\pi\sqrt{2} m^{3/2} \varepsilon^{1/2} d\varepsilon \\
 \Omega(\varepsilon) d\varepsilon &= \frac{\int d^3q \cdot \int d^3p}{h^3} \\
 &= \frac{V \cdot 4\pi p^2 dp}{h^3} \\
 &= \frac{V}{h^3} \cdot 4\pi \cdot 2m\varepsilon \cdot \sqrt{2m} \frac{1}{2\sqrt{\varepsilon}} d\varepsilon \\
 \omega(\varepsilon) &\sim \varepsilon^{1/2} d\varepsilon = g(\varepsilon) d\varepsilon
 \end{aligned}$$

Density of states $(g(\varepsilon) = \varepsilon^{1/2}) = g_i$ (degeneracy at Energy ε).

{no. of microstates per unit energy interval at energy ε }

for n-D : $V \propto \int p^{n-1} dp$ ($\Omega \propto \varepsilon^{n/2}$) for $(n \geq 2)$.

case

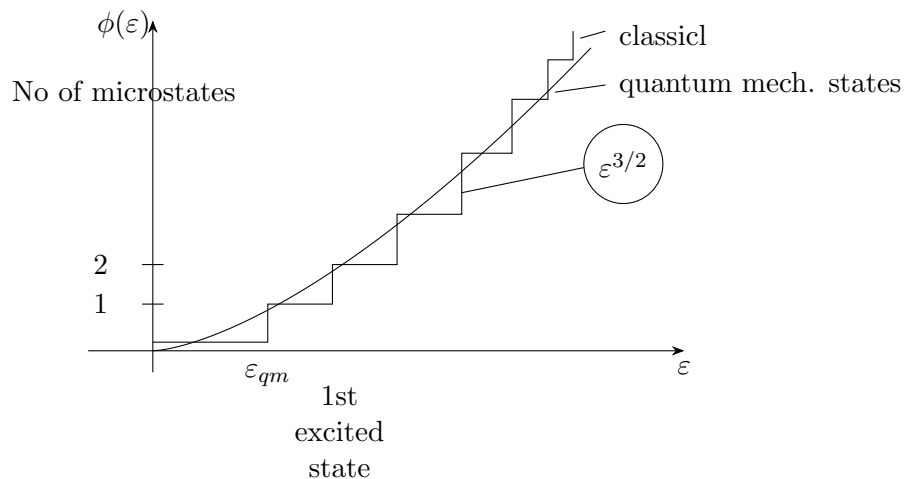
$$\begin{aligned}
 \text{for 2-D container } dv &= 2\pi p dp = 2\pi\sqrt{2m\varepsilon} \frac{2m}{2\sqrt{\varepsilon}} \\
 &= \text{Konstant}
 \end{aligned}$$

density of states becomes constant. in 2-D

case for 1-D

$$\phi(\varepsilon) = \frac{d}{d\varepsilon} \left(\frac{V}{h^3} \varepsilon^{1/2} \right) = \frac{1}{2} \varepsilon^{-1/2} \frac{V}{h^3}$$

Density of states $\uparrow$ as $\varepsilon \downarrow$



As $\varepsilon \uparrow$ no of states increases [3-D case]

(microscopic)

Now, we want to find $\Omega(E)$ so we will like to assume again that particles are free and

$$E = E_1 + E_2 + E_3 + \cdots E_N$$

Sum of energy of individual particles = total energy

For any two particles with '3' states, together they occupy $3^2 = 9$ states.

$$\Omega(E) \propto \varepsilon^{\alpha N} \propto E^{\alpha N}$$

{ if one particle can take $\omega(\varepsilon)$ states = $k\varepsilon^{1/2}$ with energy ε & $\varepsilon + d\varepsilon$

independent N particles will take — $k\varepsilon^{(1/2)N}$ }

if all ε_i are of same order of magnitude ε } approximation

$$E = N\varepsilon$$

So, $\Omega(E) \propto \varepsilon^{\alpha N} \propto \left(\frac{E}{N}\right)^{\alpha N}$.

for very large N , $\Omega(E)$ is astronomically large no.

$$\alpha = \frac{3}{2} \text{ or } \frac{1}{2} \text{ or } \frac{5}{2} \cdots$$

Lecture 22 (time 30:15)

Note

* Most probable macro state i.e. $P(E)$ is maximum = {eq^m state}

So the system which has largest no of microstates, that macrostate is the most probable state.

For Most probable macrostate it implies;

$$\frac{\partial P(E)}{\partial E} = 0 \quad P(E) = \frac{\Omega(E) \Omega'(E')}{\Omega_{Tot}(E_{Tot})}$$

as $P(E)$ is very large, let's take $\log P(E)$ & differentiate.

$$\frac{\partial}{\partial E} (\ln P(E)) = 0$$

Probability

$$dP = \frac{\partial P}{\partial E} dE + \frac{\partial P}{\partial V} dV + \frac{\partial P}{\partial N} dN$$

for eq^m

$$\left(\frac{\partial P(E)}{\partial E} = \frac{\partial P}{\partial V} = \frac{\partial P}{\partial N} = 0 \right)$$

$$\frac{\partial (\ln \Omega(E))}{\partial E} + \frac{\partial}{\partial E} (\ln \Omega'(E')) = 0$$

$$E + E' = E_{tot}$$

$$E' = E_{tot} - E \quad \Rightarrow \quad dE' = -dE$$

there fore

$$\boxed{\frac{\partial \ln(\Omega(E))}{\partial E} = \frac{\partial \ln \Omega'(E')}{\partial E'}}$$

condⁿ for thermal eq^m / most probable macrostate.

A, B two different kind of matter fluid, (gas - liquid).

$$\frac{\partial}{\partial E} (\ln \Omega(E)) = \frac{\partial}{\partial E'} (\ln \Omega'(E')) = [E^{-1}] \rightarrow \text{dimensions}$$

or,

$$\text{some property of } A = \text{some property of } B = \frac{1}{k_B \text{Temp}} \quad \} \underline{\text{Imp.}}$$

defⁿ of temperature

$$(dE = TdS - Pdv + \mu dN$$

$$T = \left. \frac{\partial E}{\partial S} \right|_{V, N}; S = k_B \ln \Omega \Bigg).$$

lets define fun. $\beta(E)$ so for this property of A/B .

$$\boxed{\beta(E) = \frac{\partial \ln \Omega(E)}{\partial E}} = \frac{1}{k_B T(E)}$$

By defⁿ in equilibrium

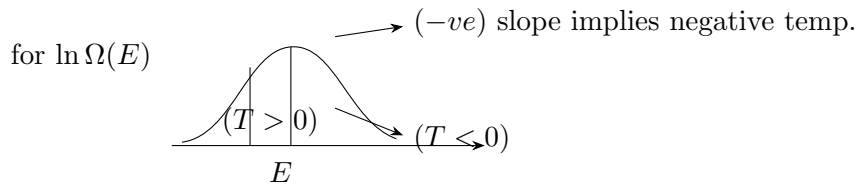
$$\boxed{T_A = T_B}$$

“ Any two parts have same temp in large isolated system under thermal equilibrium”. (large d.o.f).

“ ofcourse each part must be large enough to talk about temp, as temp. is not defined for a single particle”.

If $\Omega(E)$ is not strictly increasing function and if so

$$\frac{\partial}{\partial E} \ln(\Omega(E)) < 0 \Rightarrow \text{negative temperature.}$$



So possibility of $(-ve)$ temp is built in our defⁿ of temp.

$$\text{since } \Omega = \Omega(N, V, E) \quad \text{from } \left(\frac{\partial \ln \Omega}{\partial V} = 0 \right) \& \left(\frac{\partial \ln \Omega}{\partial N} = 0 \right)$$

$$\frac{\partial \ln \Omega}{\partial V} = \frac{\partial \ln \Omega'}{\partial V'} \equiv \left(\frac{P_A}{T_A} = \frac{P_B}{T_B} \right) \Rightarrow (P_A = P_B)$$

$$\frac{\partial \ln \Omega}{\partial N} = \frac{\partial \ln \Omega'}{\partial N'} \Rightarrow \left(\frac{\mu_A}{T_A} = \frac{\mu_B}{T_B} \right) \Rightarrow (\mu_A = \mu_B)$$

(in thermodynamic terms)

$$\begin{aligned} E &= TS - PV + \mu N \\ S &= \frac{E}{T} + \frac{PV}{T} - \frac{\mu N}{T} \\ \left(\frac{\partial S}{\partial N} = \frac{-\mu}{T} \right) ; \left(\frac{\partial S}{\partial V} = \frac{P}{T} \right) \end{aligned}$$

Entropy

$$S(E, V, N) = k_B \ln (\Omega(E, V, N))$$

Thermodynamics in Nutshell :-

1st law of thermodynamics :-

$$dQ = dU + dW$$

dQ = not a perfect differential

$$\boxed{du = dQ - dW}$$

chemical pot.

$$= dU + Pdv - \mu dN \quad \rightarrow \text{Generalized fluxes.}$$

$$dQ = dU - \sum F_i dx_i$$

$\hookrightarrow$ generalized forces

$$E \leftrightarrow U$$

Entropy representation of microcanonical ensemble $S = S(E, V, N)$

$$\boxed{dS = \frac{dE}{T} + \frac{Pdv}{T} - \frac{\mu dN}{T}}$$

tells us

$$\left(\frac{\partial S}{\partial E} \right)_{V,N} = \frac{1}{T}$$

(Energy representation) $\Rightarrow dE = TdS - Pdv + \mu dN \Rightarrow E = E(S, V, N)$

9 Lecture 23

‘E’ is some kind of thermodynamic potential

where, $E = E(S, V, N)$

E = Energy

$$\begin{aligned} T &= \left(\frac{\partial E}{\partial S} \right)_{V,N} \\ P &= - \left(\frac{\partial E}{\partial V} \right)_{S,N} \\ \mu &= \left(\frac{\partial E}{\partial N} \right)_{S,V} \end{aligned}$$

$$\Leftarrow (dE = TdS - Pdv + \mu dN)$$

What is significance of above eqⁿ set ?

we know that $E(S, V, N)$'s' is not a control variable

So lets change that by legendre transformations.

$$E = E(S, V, N)$$

↓

(Internal Energy)

Assumption

" E is homogeneous function of thermodynamic variables $\underline{S, V, N}$ ".

for that we have to assume we are in thermodynamic limit of thermodynamic system i.e. $V \rightarrow \infty, N \rightarrow \infty$ & $\frac{N}{V} = S$ (fixed).

then S, V, N which increase with system size (extensive quantities) which leads to some E .

then we can assume E is homogeneous fun of S, V, N of degree $\underline{1}$.

Euler's theorem

for any homogenous function of degree 'r'

$$f = f(x, y, z \dots)$$

i.e. $f(\lambda x, \lambda y, \lambda z, \dots) = \lambda^r f(x, y, z, \dots)$.

or

$$\left(x \frac{\partial f}{\partial x} + y \frac{\partial f}{\partial y} + z \frac{\partial f}{\partial z} + \dots \right) = r f$$

Then for $E = E(S, V, N)$

$$S \left. \frac{\partial E}{\partial S} \right|_{V, N} + V \left. \frac{\partial E}{\partial V} \right|_{S, N} + N \left. \frac{\partial E}{\partial N} \right|_{S, V} = E$$

$$S(T) + V(-P) + N(\mu) = E \quad (\text{Euler's relations})$$

So E must be equal to $TS - PV + \mu N$.

or

$$E - TS + PV = \mu N$$

we also know that $G = E - TS + PV$

so $G = \mu N$

$$\text{So } \Rightarrow \mu = \frac{G}{N} = \text{Gibbs free energy per particle}$$

let us find dE

$$dE = (TdS - Pdv + \mu dN) + (SdT - vdP + Nd\mu)$$

But laws of thermodynamics tells us that,

$$(dE = TdS - Pdv + \mu dN) \text{ only}$$

So it implies that,

$$SdT - v dP + N d\mu = 0$$

$$([\text{?}]) \Rightarrow d\mu = \left(\frac{V}{N}\right) dP - \left(\frac{S}{N}\right) dT$$

$$= v dP - s dT$$

volume per particle $\leftarrow$ specific volume per particle

s = entropy per particle $\rightarrow$ specific entropy (per particle)

So, $d\mu = v dP - s dT$ $\rightarrow$ Gibbs–Duhem relation

it implies, $\boxed{\mu = \mu(P, T)}$

So once we know (P, T) we know the chemical potential (μ)

If we assume ‘T’ is some kind of generalised force & ds as some generalised flux

$$\text{then, } dE = (T ds) + \sum_i F_i dX_i$$

$$dE = \sum_i F_i dX_i$$

So homogeneity arguments tells

$$E = \sum_i F_i X_i \Rightarrow \left(\sum_i X_i dF_i = 0 \right).$$

$\downarrow$

(Generalised Duhem relation)

This comes from our assumption of E to be a homogeneous function of degree 1.

If E is not homogeneous function.

Legendre transforms

$$E(S, V, N) \longrightarrow H(S, P, N)$$

(Enthalpy)

$$H = E + PV$$

$$dH = dE + P dv + v dP$$

$$= T ds - P dv + \mu dN + P dv + v dP$$

$$dH = T ds + v dP + \mu dN$$

So $H = H(S, P, N)$

$$E(S, V, N) \longrightarrow F(T, V, N)$$

$$F = E - TS$$

$$dF = T dS + \mu dN - P dv - T dS - S dT$$

$$dF = -S dT - P dv + \mu dN \quad \Rightarrow F = F(T, V, N)$$

$$E(S, V, N) \longrightarrow G(T, P, N)$$

$$G = E - TS + PV. \quad (\text{Gibbs free energy})$$

$$E(S, V, N) \longrightarrow \Phi_1(S, V, \mu)$$

$$\Phi_1 = E - \mu N$$

$$E(S, V, N) \longrightarrow \Phi_2(S, P, \mu)$$

$$\Phi_2 = E + PV - \mu N$$

$$E(S, V, N) \longrightarrow \Phi_3(T, V, \mu)$$

$$\Phi_3 = E - TS - \mu N$$

But we know $E = TS - PV + \mu N$

$$\Rightarrow E - TS - \mu N = -PV$$

So $\boxed{\Phi_3 = -PV} \rightarrow$ “grand potential”

↓

The reason is that this potential Φ_3 is connected to grand canonical ensemble but others above mentioned potentials are only linked to Canonical ensemble of stat. mech.

In,

$$E = TS - PV + \mu N$$

$(T, P, \mu) \rightarrow$ Intensive variables

$(S, V, N) \rightarrow$ Extensive ”.

<u>Field variables (Intensive)</u>	<u>State variables. (extensive)</u>
T	S
P	V
μ	N
Stress σ_{ij}	Strain $\epsilon_{ij}(= q_i)$
$\vec{E}$	$P \rightarrow$ polarisation per unit volume
$\vec{H}$	M ; magnetic dipole moment per unit volume

$$dF = -s dT - P dv + \mu dN \quad \text{and} \quad ; \quad dG = -s dT + v dP + \mu dN$$

lets look at these 2 relations,

$$S = - \left(\frac{\partial F}{\partial T} \right)_{V, N}$$

So,

$$\left(\frac{\partial F}{\partial T} \right)_{V, N} = \left(\frac{\partial G}{\partial T} \right)_{P, N}$$

Similarly,

$$P = - \left(\frac{\partial F}{\partial V} \right)_{T, N}$$

$$\mu = \left(\frac{\partial F}{\partial N} \right)_{T, V}$$

It says these thermodynamic potentials ($U, G, F, H \dots$), their 1st partial derivative with their individual variables on which they depend are other thermodynamic variables.

$$\begin{aligned}
dQ &= dU + dW \\
TdS &= dE + P dv - \mu dN \\
\left(\frac{dQ}{dT}\right)_{V,N} \text{ or } \left(\frac{\partial Q}{\partial T}\right)_{V,N} &= C_V = T \left(\frac{\partial S}{\partial T}\right)_{V,N} = \left(\frac{\partial E}{\partial T}\right)_{V,N} \quad +0-0 \\
C_V &= T \left(\frac{\partial S}{\partial T}\right)_{V,N} \quad \text{but,} \quad S = - \left(\frac{\partial F}{\partial T}\right)_{V,N}
\end{aligned}$$

So

$$C_V = -T \left(\frac{\partial^2 F}{\partial T^2}\right)_{V,N}$$

response functions are 2nd derivative of thermodynamic potentials

But we know C_V can not be negative } lesea teliers principle.

So,

$$_V \left(\frac{\partial^2 F}{\partial T^2} < 0\right).$$

$$C_P = T \left(\frac{\partial S}{\partial T}\right)_{P,N} = -T \left(\frac{\partial^2 G}{\partial T^2}\right)_{P,N} \quad \text{as,} \quad \left(S = - \left(\frac{\partial G}{\partial T}\right)_{P,N}\right).$$

also

$$(C_P > C_V) \quad \text{so,} \quad \left|\frac{\partial^2 G}{\partial T^2}\right| > \left|\frac{\partial^2 F}{\partial T^2}\right|.$$

Isothermal compressibility (fixed T,N)

$$K_T = \frac{1}{\text{bulk modulus}} = \frac{-\Delta v}{V \Delta P} = -\frac{1}{v} \left(\frac{\partial v}{\partial P}\right)_{T,N} \quad (\text{as } P \uparrow \quad v \downarrow)$$

Since $\left(V = \left(\frac{\partial G}{\partial P}\right)_{T,N}\right)$

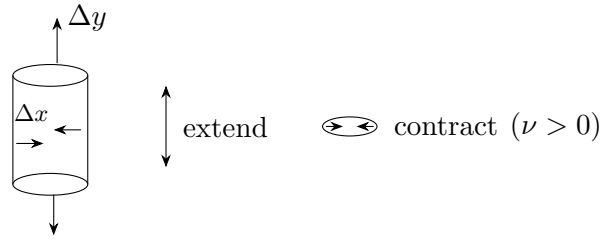
$$\left(\frac{\partial v}{\partial P}\right)_{T,N} = \left(\frac{\partial^2 G}{\partial P^2}\right)_{T,N}$$

$$K_T = -\frac{1}{v} \left(\frac{\partial^2 G}{\partial P^2}\right)_{T,N}$$

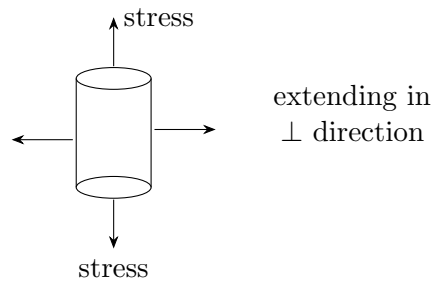
Poisson's ratio (ν) :-

Poisson ratio is measure of Poissons effect, that describes the expansion or contraction of a material in direction perpendicular to the direction of loading

$$\left(\nu = \frac{\Delta x}{\Delta y} = \frac{\text{contraction strain}}{\text{extension strain}}\right)$$



It's possible that ($\nu < 0$) such that upon extending in one direction object/material also expand in $\perp$ dir.



It turns out $\boxed{-1 < \nu \leq \frac{1}{2}}$

For ideal gas

$$\begin{aligned}
 PV &= nRT \\
 P dv + v dP &= nR dT \\
 v dP &= -P dv \\
 \frac{dv}{v} &= -\frac{dP}{P} \\
 \Rightarrow K_T &= -\frac{1}{v} \frac{\partial v}{\partial P} = -\frac{1}{V} \cdot \left(\frac{-v}{P} \right) = \frac{1}{P}
 \end{aligned}$$

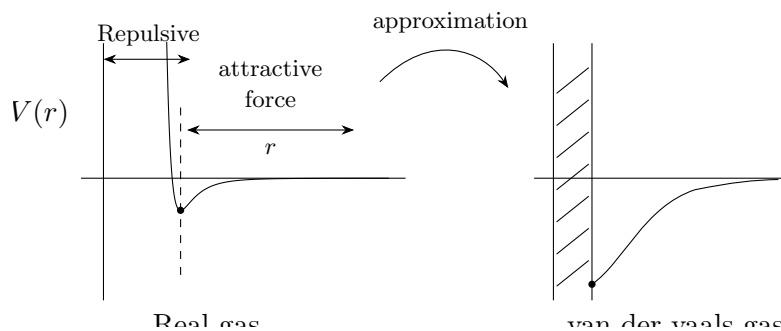
If K_T is high pressure is low.

for adiabatic process

$$\begin{aligned}
 P v^\gamma &= \text{const} \\
 dP v^\gamma + P \gamma P^{\gamma-1} dV &= 0 \\
 v^{\gamma-1} (v dP + \gamma P dv) &= 0 \\
 \Rightarrow K_T &= -\frac{1}{v} \frac{\partial v}{\partial P} = +\frac{1}{v} \cdot \frac{v}{\gamma P} = \frac{1}{\gamma P} \\
 \left(K_{S,N} &= \frac{1}{\gamma P} \right).
 \end{aligned}$$

What happens if we have van der vaals gas :-

$$P = \frac{nRT}{V - bn} - \frac{an^2}{V^2}$$



{This repulsion arose from pauli exclusion principle

$$V(r) = V_0 \left[\left(\frac{a}{r} \right)^{12} - \left(\frac{a}{r} \right)^6 \right] \quad \text{"6-12" potential} \quad \left(\begin{array}{l} \text{lennard-Jonnes} \\ \text{potential} \end{array} \right)$$

$\left(\frac{a}{r} \right)^6 \rightarrow$ attractive force potential

$\left(\frac{a}{r} \right)^{12} \rightarrow$ need quantum mech effect. (repulsive term).

Attractive force term arise from dipole moment attraction.



Random fluctuations can cause dipole moment.

interaction energy of any of dioples goes like

$$(U) \sim \vec{p} \cdot \vec{E}$$

greater the field greater the seperation

So

$$\vec{p} = \alpha \vec{E}$$

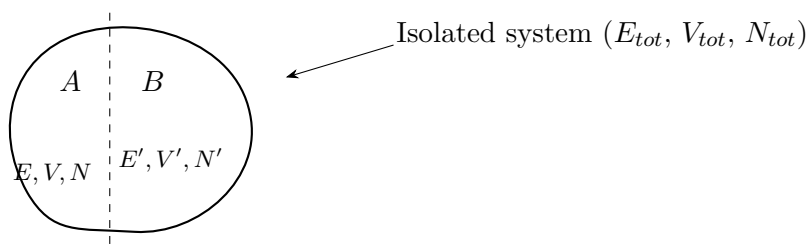
$$U \sim \alpha |E|^2 \quad \vec{E} \text{ of dipole} = \frac{2kp}{r^3}$$

$$U \propto E^2 \propto \frac{1}{r^6}$$

This formula holds true only for molecules with temperory dipole moment.

10 Lecture 24

Let us try to understand the distribution of energy in a subsystem of a (large isolated system.



condⁿ of eqⁿ implies,

$$\frac{\partial \ln \Omega}{\partial E} = \frac{\partial \ln \Omega'}{\partial E'} = \frac{1}{k_B T}$$

$$(T_A = T_B).$$

Probability that subsystem A has energy E

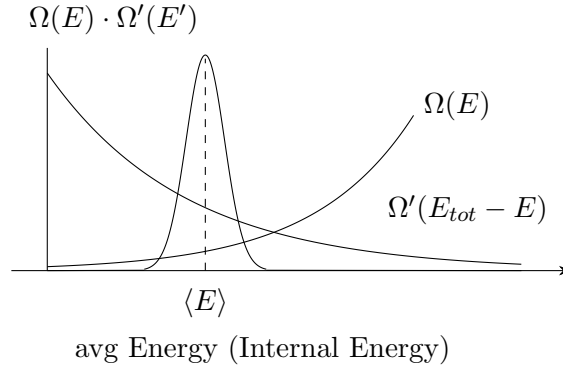
$$P(E) = \frac{\text{No of microstates of total system such that A has energy 'E'}}{\text{Total no of accessible microstates of system with energy } E_{tot}}.$$

$$P(E) = \frac{\Omega(E) \cdot \Omega'(E')}{\Omega_{tot}(E_{tot})}$$

where $(0 \leq E \leq E_{tot})$.

We know that

$$\Omega'(E') = \Omega'(E_{tot} - E) \propto (E_{tot} - E)^N$$



$$\ln P(E) = \ln \Omega(E) + \ln \Omega'(E_{tot} - E) - \ln \Omega_{tot}(E_{tot})$$

let us expand $\ln P(E)$ about some point $\bar{E}$ or $\langle E \rangle$.

$$\ln P(E) \approx \ln P(\bar{E}) + (E - \bar{E}) \left. \frac{\partial \ln P(E)}{\partial E} \right|_{(E=\bar{E})}^{=0} + \frac{(E - \bar{E})^2}{2!} \left. \frac{\partial^2 \ln P(E)}{\partial E^2} \right|_{E=\bar{E}}$$

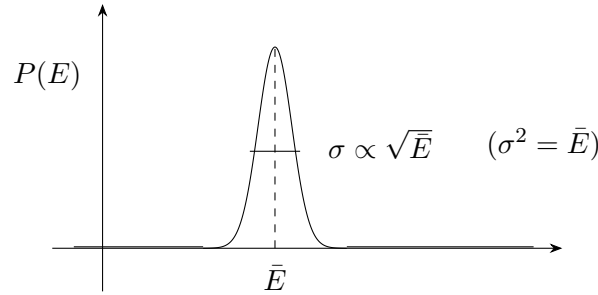
Since $P(E)$ is max at $E = \bar{E}$, $\left(\frac{\partial \ln P(E)}{\partial E} \right)_{\bar{E}} = 0$

$$\ln P(E) = \ln P(\bar{E}) + \frac{(E - \bar{E})^2}{2!} \left. \frac{\partial^2 \ln P(E)}{\partial E^2} \right|_{\bar{E}}$$

Since slope of slope at $\bar{E}$ is negative.

$$P(E) = P(\bar{E}) \cdot e^{-K \frac{(E - \bar{E})^2}{2!}}$$

$$P(E) = (\text{Normalisation const}) \cdot e^{-\alpha \frac{(E - \bar{E})^2}{2\bar{E}}} \quad \text{'}\alpha\text{' to make dimensionless.}$$



So relative fluctuations. goes like $\frac{\sqrt{\bar{E}}}{\bar{E}} = \frac{1}{\sqrt{\bar{E}}}$

$$\left(\bar{E} = N \frac{f}{2} k_B T \right)$$

relative fluctuations about mean $\propto \frac{1}{\sqrt{N}}$

So for ($N \approx 10^{24}$) fluctuations in eq^m about mean value in avg Energy $\propto 10^{-12}$. which is negligible. That is why in thermodynamics we only talk about $\langle \bar{E} \rangle$ or $\langle E \rangle$ rather than $\langle E_{tot} \rangle$.

$$\begin{aligned} \ln P(\varepsilon) &= \ln \Omega'(E') - \ln \Omega_{tot}(E_{tot}) \\ &= \ln \Omega'(E_{tot} - E) - \ln \Omega_{tot}(E_{tot}) \end{aligned}$$

Since ($E_{tot} \gg \varepsilon$) let us expand $\ln \Omega'(E')$ about (E_{tot}).

$$\ln P(\varepsilon) = \left(\ln \Omega'(E_{tot}) - \varepsilon \left. \frac{\partial \ln \Omega'(E')}{\partial E'} \right|_{E'=E_{tot}} + \dots \right) - \ln \Omega_{tot}(E_{tot})$$

$$P(\varepsilon) = (\text{const}) e^{-\beta \varepsilon} \quad \left| \quad \beta = \frac{\partial \ln \Omega'(E')}{\partial E'} = \text{inverse temp} = \frac{1}{k_B T} \text{ of heat bath} \right.$$

to find (const) we need to know $\Omega'(E_{tot})$ (info about heat bath). but we don't need that too, we can normalize $P(\varepsilon)$ and find normalisation const.

$$P(\varepsilon) = \frac{e^{-\beta \varepsilon}}{\sum_{\varepsilon} e^{-\beta \varepsilon}}$$

So Normalisation const. is fixed by properties of subsystem alone and we eliminate whatever happens outside"

for a moment, let us assume (A) has discrete energy levels.

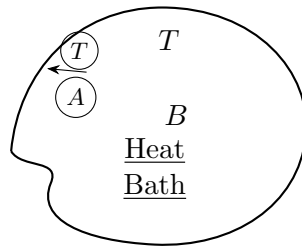
$$\left. \begin{array}{l} \text{degenerate} \\ \text{states} \end{array} \right\} \begin{array}{l} \text{---} \text{---} \text{---} \text{---} \varepsilon_3 \\ \text{---} \text{---} \varepsilon_2 \\ \\ \text{---} \varepsilon_1 \\ \text{---} \varepsilon_0 \end{array}$$

$$P(\varepsilon_i) = \frac{e^{-\beta\varepsilon_i}}{\sum_j e^{-\beta\varepsilon_j}} \quad \rightarrow \text{summed over all states not levels.}$$

The only information about heat bath to be known is its temp. unlike (case of microcanonical ensamble where $E = \bar{E}$ at eq^m) The total energy of system in thermal eqⁿ with heat bath is not fixed. It is fluctuating with in diff energy with $P(\varepsilon_i) = e^{-\beta\varepsilon_i} / \sum_i e^{-\beta\varepsilon_j}$ but the avg. value of $\langle \varepsilon_i \rangle$ is independent of time, because $P(\varepsilon_i)$ is independent of time

The factor $\left(\sum_{\text{states } j} e^{-\beta\varepsilon_j} \right)$ is called ‘Z’ the canonical partition function.

and our system

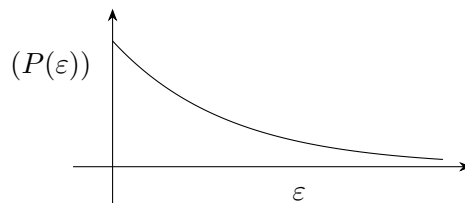


in Thermal-eq^m at temp T is called “canonical ensamble”.

$$\left(Z = Z(T, V, N) = \sum_{\text{states } j} e^{-\beta\varepsilon_j} = \sum_{\text{levels } i} g_i e^{-\beta\varepsilon_i} \right)$$

The subsystem “A” in thermal eq^m of this kind is not in fixed energy state. It has got probability distributions for certain energy values. For a quantum system “A” is not even in pure energy eigen state. It will be in superposition of all energy eigen states. {mind blowing fact.

$$(P(\varepsilon) \propto e^{-\beta\varepsilon_i}).$$



distribution of energy goes like exponential decay rather than steep gaussian distribution.

β = property of bath.

In terms of energy levels,

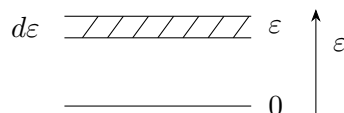
$$Z = \sum_{\text{levels } j} (g_j) e^{-\beta\varepsilon_j}$$

$\rightarrow$ degeneracy of level ε_j .

for continuous energy levels,

$$P(\varepsilon) = \frac{e^{-\beta\varepsilon}}{\int_0^\infty e^{-\beta\varepsilon} g(\varepsilon) d\varepsilon}.$$

now instead of counting degeneracy (g_j) we ask



how many states with energy ε & $\varepsilon + d\varepsilon$

$$= g(\varepsilon) d\varepsilon$$

$\hookrightarrow$ density of states

$g(\varepsilon)$ = no of states in $d\varepsilon$ range per unit energy ε

$$Z(T, V, N) \equiv Z = \int_0^\infty e^{-\beta\varepsilon} g(\varepsilon) d\varepsilon \quad (\text{general canonical partition fun.})$$

So,

$$p(\varepsilon) = \frac{e^{-\beta\varepsilon}}{\int_0^\infty e^{-\beta\varepsilon} g(\varepsilon) d\varepsilon}.$$

So Z is Laplace transform of $g(\varepsilon)$, in which transform variable is β , instead of 's'. (Laplace transform or Moment generating functions for continuous variables)



$$\left. \begin{aligned} g(\varepsilon) &\longrightarrow \text{system properties} \quad (\text{mech. aspect classical/quantum}). \\ e^{-\beta\varepsilon} &\longrightarrow \text{statistical aspect.} \end{aligned} \right\}$$

$$e^{-\beta\varepsilon} g(\varepsilon) \longrightarrow \underline{\text{stat. mech.}}$$

(Q) What is average energy?

$$\begin{aligned} \langle E \rangle &= \sum_{(\text{states } j)} P(E_j) E_j \\ &= \frac{\sum_{(\text{states } i)} E_i e^{-\beta E_i}}{Z(T, V, N)} \\ &= \frac{\sum_{(\text{states } i)} E_i e^{-\beta E_i}}{\left(\sum_{(\text{states } j)} e^{-\beta E_j} \right)} = \frac{1}{Z} \left(\sum_{(\text{states } i)} E_i e^{-\beta E_i} \right). \end{aligned}$$

$$\boxed{\langle E \rangle = -\frac{\partial(\ln Z)}{\partial\beta}} \quad \left(\frac{-1}{Z} \cdot \frac{\partial Z}{\partial\beta} = +\frac{1}{Z} \sum E_i e^{-\beta E_i} \right)$$

$$= -\frac{1}{Z} \frac{\partial Z}{\partial\beta}$$

So, ($C_v \geq 0$) $\therefore$

In thermodynamics we only deal with $\langle E \rangle$. So we can't calculate (C_v) or $\underline{C_p}$. So C_p, C_v are input in Thermo-dynamics.

11 Lecture 25: Connection with thermodynamics

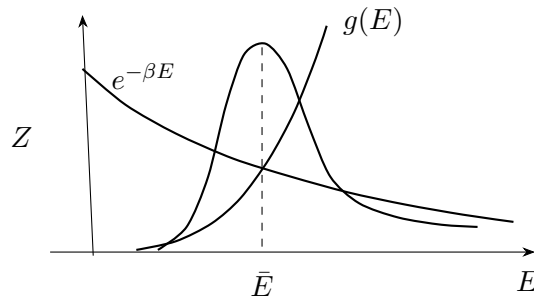
Recall that the canonical partition function of small subsystem in heat bath of temp. $T = \frac{1}{k_B\beta}$

$$Z = \sum_{\text{states } j} e^{-\beta E_j} \longrightarrow \int_0^\infty dE e^{-\beta E} g(E)$$

$$g(E) \propto E^{1/2}$$

$$g(E) \propto E^{\frac{N}{2}} \text{ or } \propto E^N \quad \text{for } \underline{N = 10^{24}}.$$

Lets plot (Z vs E)



$$Z = \int_0^\infty dE e^{-\beta E} g(E) = e^{-\beta \bar{E}} \Omega(\bar{E})$$

$$(?) \text{ maybe } \sim (\text{by me}) \quad (\Omega(E) \approx \Omega(\bar{E}) = \int_0^\infty g(E) dE)$$

$\Omega(\bar{E})$ = total no of microstates

So,

$$\ln Z = \ln \Omega(\bar{E}) - \beta \bar{E}$$

$$-k_B T \ln Z = -k_B T \ln \Omega(\bar{E}) + \bar{E}$$

But we know that entropy (S) is defined as

$$\boxed{S = k_B \ln \Omega}$$

$$\begin{aligned}
-k_B T \ln Z &= -TS + \bar{E} \quad (\bar{E} = \langle E \rangle) = U \text{ (Internal energy)} \\
&= F/A \text{ (Hemholtz free energy).} \\
&\Rightarrow \boxed{Z = e^{-\beta F}}
\end{aligned}$$

So, the whole idea was to write

$$Z = \sum_{\text{states } j} e^{-\beta E_j} \quad \text{in terms of} \quad e^{-\beta E_{\text{eff}}}$$

$\hookrightarrow$ effective energy

$$(Z = e^{-\beta F}). \quad (F = \text{Helmholtz free energy}).$$

It needs coherent punch from bath molecules to provide maximum energy to subsystem 'A'. So it is less likely to get high energy from a heat bath; that is the reason the energy distribution goes like

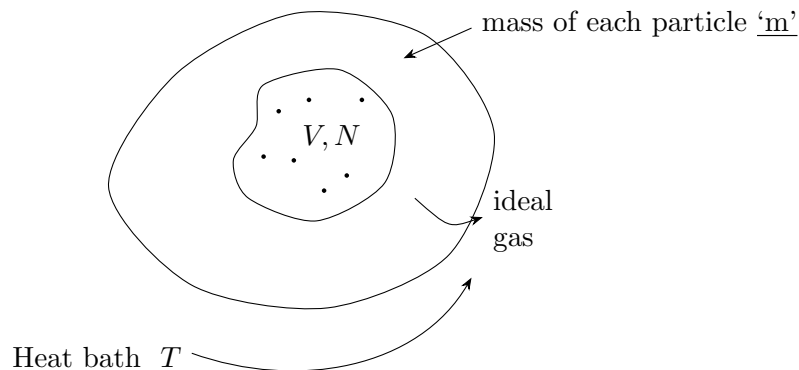
$$P(E_i) = \frac{e^{-\beta E_i}}{\sum_{\text{states } j} g_j e^{-\beta E_j}}$$

Let us find Pressure:

We know that

$$\begin{aligned}
F &= U - TS \\
dF &= dU - TdS - SdT \\
&= TdS - PdV + \mu dN - TdS - SdT \\
dF &= -SdT - PdV + \mu dN \\
P &= -\left(\frac{\partial F}{\partial V}\right)_{T,N}
\end{aligned}$$

(Partition function for classical ideal gas):



Assume, All particles are independent of each other and there is no interaction b/w the particles.

$$Z = \sum_{\substack{\text{states} \\ \text{of} \\ \text{gas}}} e^{-\beta E} = \sum e^{-\beta(\varepsilon(1)+\varepsilon(2)+\varepsilon(3)+\dots)}$$

where $\varepsilon(i)$ is energy of i th particle of whole system.

If all particles are independent of each other, then partition function factorize into –

$$Z = \sum e^{-(\beta\varepsilon + \beta\varepsilon + \beta\varepsilon + \dots)} = \sum_{\text{states of gas}} \left(e^{-\beta\varepsilon} \cdot e^{-\beta\varepsilon} \cdot e^{-\beta\varepsilon} \dots \right)$$

$$Z = \left(\sum_{\text{states of any particle}} e^{-\beta\varepsilon} \right)^N$$

→ Only true if particles are distinguish-able.

$$Z = \left(\int_0^\infty e^{-\beta\varepsilon} \underbrace{g(\varepsilon)d\varepsilon}_{\Omega(E)} \right)^N = \left(\frac{1}{h^3} \left(\int_V d^3r \int d^3p e^{-\frac{\beta p^2}{2m}} \right) \right)^N$$

$$Z = \left(\int_0^\infty e^{-\beta E} \Omega(E) dE \right)^N$$

$$= \left(\frac{V}{h^3} \left(\int e^{-\frac{\beta p_x^2}{2m}} dp_x \int e^{-\frac{\beta p_y^2}{2m}} dp_y \int e^{-\frac{\beta p_z^2}{2m}} dp_z \right) \right)^N$$

$$= \left(\frac{V}{h^3} \left((2\pi m k_B T)^{1/2} \cdot (2\pi m k_B T)^{1/2} (2\pi m k_B T)^{1/2} \right) \right)^N$$

$$Z = \left(\frac{V}{h^3} (2\pi m k_B T)^{3/2} \right)^N$$

$$\Omega(\varepsilon) = \frac{1}{h^3} \int_V d^3r \int_{-\infty}^\infty d^3p \quad \left(\int_{-\infty}^\infty e^{-ax^2} dx = \sqrt{\frac{\pi}{a}} \right)$$

$$Z = e^{-\beta F}$$

$$\ln Z = -\beta F \Rightarrow \left(F = \frac{-1}{\beta} \ln Z \right) = -k_B T N \ln \left(\frac{V (2\pi m k_B T)^{3/2}}{h^3} \right)$$

$$P = - \left(\frac{\partial F}{\partial V} \right)_{T,N} = \frac{1}{\beta} \cdot \frac{1}{Z} \cdot \frac{\partial Z}{\partial V} \Big|_{T,N}$$

$$= \frac{1}{\beta Z} \cdot \frac{1}{h^3}$$

$$= N k_B T \cdot \frac{1}{V \left(\frac{(2\pi m k_B T)^{3/2}}{h^3} \right)} \cdot \frac{(2\pi m k_B T)^{3/2}}{h^3}$$

$$P = \frac{N k_B T}{V} \Rightarrow \boxed{PV = N k_B T}$$

When we will do Q-M ; $E = pc$ for photons in black body cavity at temp T . then

$$\bar{E} \neq \frac{3}{2} k_B T \quad (\text{even in 3-D}).$$

$\hookrightarrow E = p^1 c$ So

$$\underline{\bar{E} \propto 3k_B T^4 f(\cdot)}$$

$\hookrightarrow$ Bose statistics.

So,

$$PV = Nk_B T \quad ; \quad U = \frac{3}{2} Nk_B T$$

$$U = \frac{3}{2} \cdot PV$$

$$\Rightarrow \boxed{PV = \frac{2U}{3}}$$

$\rightarrow$ avg energy density.

$$\text{or, } \left(P = \frac{2u}{3} \right) \quad \left(u = \frac{U}{V} = \text{avg Energy per unit volume} \right).$$

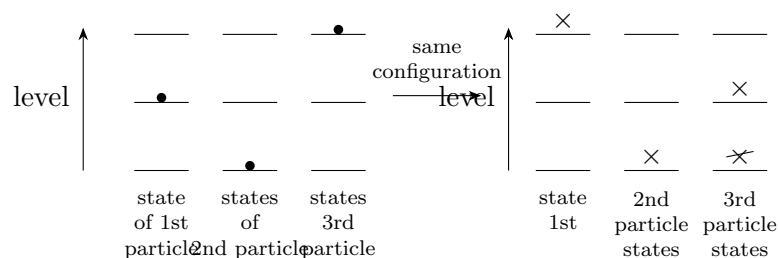
This relation is even true for quantum gases. (quantum ideal gas).

(Q) Is it correct that

$$Z = \sum e^{-\beta E} = \left(\sum e^{-\beta \epsilon} \right)^N ?$$

summed over all states of gas summed over states of any particle

No, its not true (always) here we have overcounted the number of states, since the particles are indistinguishable. (physical fact).



since particles are indistinguishable, we have overcounted the number of states.

In quantum statistics we will correct this mistake.

Instead of asking in which level particle 1 is.. and 2 is.. we ask how many particles are in ground state, 1st excited state. and so on....

So in nutshell in quantum statistics we do row wise counting instead of column wise counting performed in classical statistics. From here idea of occupation no. arose in quantum statistics.

$$dF = -SdT - PdV + \mu dN$$

$$S = - \left(\frac{\partial F}{\partial T} \right)_{V,N} \quad ; \quad e^{-\beta F} = Z$$

$$\left(F = \frac{-1}{\beta} \ln Z \right)$$

and,

$$Z = \left(\frac{V}{h^3} (2\pi m k_B T)^{3/2} \right)^N$$

$$\Rightarrow F = \frac{-N}{\beta} \ln \left(\frac{V}{h^3} \left(\frac{2\pi m}{\beta} \right)^{3/2} \right) = -Nk_B T \ln \left(\frac{V}{h^3} (2\pi m k_B T)^{3/2} \right)$$

So,

$$\begin{aligned} S &= - \left(\frac{\partial F}{\partial T} \right)_{V,N} = +Nk_B \ln \left(\frac{V}{h^3} (2\pi m k_B T)^{3/2} \right) \\ &\quad + \frac{Nk_B T}{\frac{V}{h^3} (2\pi m k_B T)^{3/2}} \cdot \frac{V}{h^3} \cdot \frac{3}{2} (2\pi m k_B T)^{1/2} (2\pi m k_B) \\ &= -Nk_B \ln \left(\frac{V}{h^3} (2\pi m k_B T)^{3/2} \right) + \frac{3}{2} Nk_B \end{aligned}$$

at $T = 0$ $\ln \left(\frac{V}{h^3} (0) \right) \rightarrow \infty$, so ideal gas is no more a good assumption for our real gas, there are interactions

and physical real gas condenses to liquid & solid which also can't be explained by assuming gas to be ideal.

Also this formula of entropy is more flawed and it is due to incorrect partition function arose from overcounting of states (also known as Gibbs paradox).

Let us find

$$\begin{aligned} \mu &= + \frac{\partial F}{\partial N} \Big|_{T,V} \quad ; \quad F = -Nk_B T \ln \left(\frac{V}{h^3} (2\pi m k_B T)^{3/2} \right) \\ \mu &= -k_B T \ln \left(\frac{V}{h^3} (2\pi m k_B T)^{3/2} \right) \end{aligned}$$

this formula is also not sensible

Even the formula for Free energy $F = -Nk_B T \ln \left(\frac{V}{h^3} ()^{3/2} \right)$ as, we expect free energy to be extensive quantity.

The 'V' in $\ln \left(\frac{V}{h^3} ()^{3/2} \right)$ is trouble here.

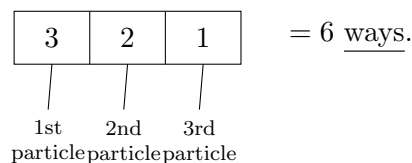
($F \propto$ any one extensive variable

$$F = Nf(\text{intensive variable}).)$$

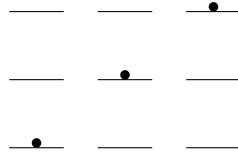
Correction to partition function :-

There are ($n!$ no of ways) particles can be arranged

ex 3 particles.



So A rough way to correct Z is to divide by ($n!$) but that is only true if no two particles attains same (state)

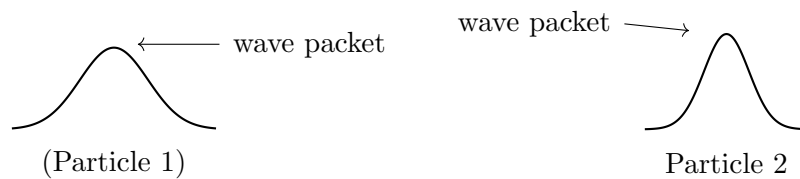


But that is only very likely if no. of states available

Since particles are quantum object, we have to move to quantum statistics for more precise understanding.

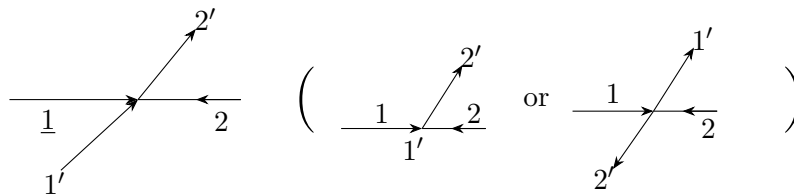
(Criterion for classical statistics to be valid) :-

In Q.M. particles are indistinguishable and x & p of particle can't be precisely defined instantaneously.



We can say that particle 1 is at this position $(x_1 \pm \Delta x_1)$ & particle 2 is at some other position $(x_2 \pm \Delta x_2)$, iff the avg distance b/w them is much larger than spread of wave packets. (i.e. non-interacting identical particles can be distinguishable).

When particles 1 & 2 interact, we can't identify them.



(particles are exchanged).

So, for classical statistics to work

$$\text{mean interparticle separation} \gg \lambda_{\text{debroglie}}$$

we still have to divide by $(N!)$ to count states correctly.

If we have N particles in volume V then volume available for each particle is $\left(\frac{V}{N}\right)$

So linear dimension available is $\left(\frac{V}{N}\right)^{1/3}$

$$\left(\frac{V}{N}\right)^{1/3} \gg \frac{h}{p_{rms}} \quad \left(\lambda_{deB} = \frac{h}{p}\right).$$

$$\left\langle \frac{p^2}{2m} \right\rangle = \frac{3}{2} k_B T \quad p_{rms} = \sqrt{3mk_B T}$$

for one particle

$$\sqrt{\langle p^2 \rangle} = p_{rms} \leftarrow = (3mk_B T)^{1/2} \text{ or } (mk_B T)^{1/2}$$

$$\left(\frac{V}{N}\right)^{1/3} \gg \frac{h}{(mk_B T)^{1/2}}$$

$$\left(\frac{V}{N}\right) \gg \frac{h^3}{(mk_B T)^{3/2}} \quad \text{or} \quad \left(\frac{1}{n}\right) \gg \frac{h^3}{(mk_B T)^{3/2}}$$

$$\boxed{1 \gg \frac{nh^3}{(mk_B T)^{3/2}}} \quad \frac{nh^3}{(mk_B T)^{3/2}} = \text{degeneracy factor}$$

So

$$\boxed{\frac{nh^3}{(mk_B T)^{3/2}} \ll 1}$$

is criteria for statistical to be valid. (n = number density of particles)

eg) Nitrogen $m \approx 10^{-26}$ kg

$$h = 10^{-34}$$

$$n = 10^{24} \text{ per cubic meter}$$

$$T = 300 \text{ K}, \quad k_B = 1.38 \times 10^{-23}$$

$$\frac{10^{24} \cdot 10^{-34 \times 3}}{(10^{-26} \cdot 10^{-23} \cdot 300)^{3/2}} = \frac{10^{-102+24}}{(10^{-51})^{3/2}}$$

$$= \frac{10^{-78}}{10^{-75}} \approx 10^{-3} \ll 1.$$

So classical statistics works.

If ($n \uparrow$) this approximation fails. (like in case of liquids/neutron stars. the degeneracy factor is of order of 10^6).

Also at low temp ($T \rightarrow 0$) this limit/condition fails and we use quantum statistics.

(Lec -26 – is of probability distributions so, It will be written in mathametical notebook of V. balki sir).

12 Lecture 27

(After) (0 : 00)

(Phase transitions)

Thermodynamics is a kind of limiting case of statistical mech. where fluctuations are neglected and so on; And these fluctuations can be taken into account if we use stat. mech. and it provides correction to thermodynamic cases, goes beyond thermostatics & calculate (C_P, C_V) which are input parameters in thermodynamics.

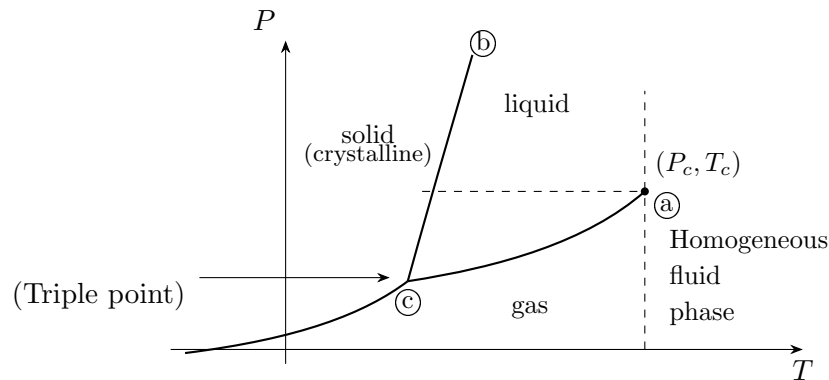
There is a place where thermodynamics fails (completely and no longer valid); it is hard problem of phase transitions of matter. In principle stat. mech. (eq^m stat. mech.) should tell us everything about system at thermal eqⁿ, including processes at phase transitions.

Thermodynamics fails at critical point (P_c, V_c, T_c).

We will use vander vaals gas to demonstrate real gases.

We have thermodynamic variables (P, V, T) out of which we only use any two of them, as third is found of using equation of state (which is different for different phases of matter – solid, liquid, gas).

Phase diagram



- (a) → liquid-gas coexisting curve or (Boiling curve)
- (b) → solid-liquid coexisting curve or melting curve
- (c) → solid-gas coexistant curve or sublimation (curve).

every point on this phase plane is assumed to at thermal eq^m state of the system.

(phase – A homogeneous thermodynamic eq^m system)

(Ice melting of glaciers is measured using rise of water level but Ice can directly convert to vapour phase, so this is a challanging problem).

(Note :- It looks like the melting curve goes on with same slope, but it does not happen, As pressure increases solid have more densely packed atoms, at very high pressure (FCC) Face centered cubic crystals formes.

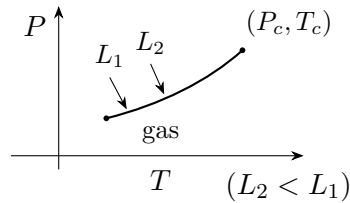
(Gibbs phase rule) :- For one kind of molecular species at most we can have only three coexisting phases. For more than one molecular species we can have more than 3 kind of phase coexisting at common point.

Critical Point (V_c, T_c, P_c) :-

The boiling curve/liquid-gas coexisting curve stops at some point on phase plane called critical point where it becomes impossible to distinguish liquid phase from gas phase.

At this point the conventional thermodynamics fails. on right of (P_c, T_c) or for $(T > T_c)$ we can't tell diff between liquid & phase. (Homogeneous fluid phase).

To convert form liquid to gas phase, we supply latent heat.



this transition is discontinuous as density of system changes discontinuously

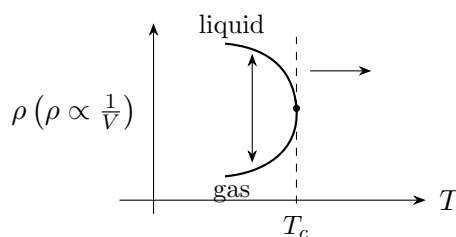
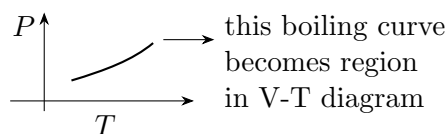
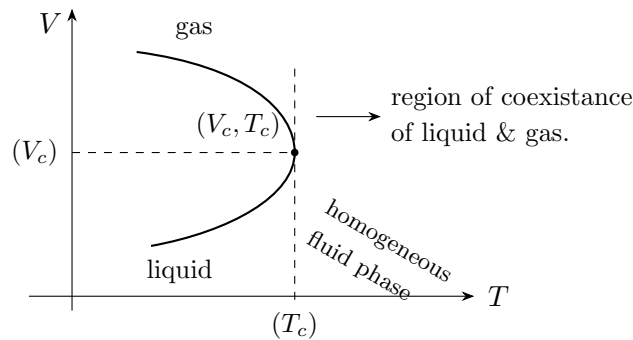
As we go up this curve (i.e. increase pressure), latent heat becomes less and less, & latent heat vanishes at critical point (i.e. we require no heat to convert from liquid to gas phase).

Also the surface tension also vanishes at critical point i.e. the meniscus (the curved upper surface of liquid) also disappears. “we have mush at critical point.”

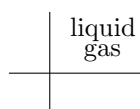
Everything that distinguish liquid from gas disappears, even the density start matching.

Lets draw melting curve in $(V - T)$ plane

volume of gas is very high

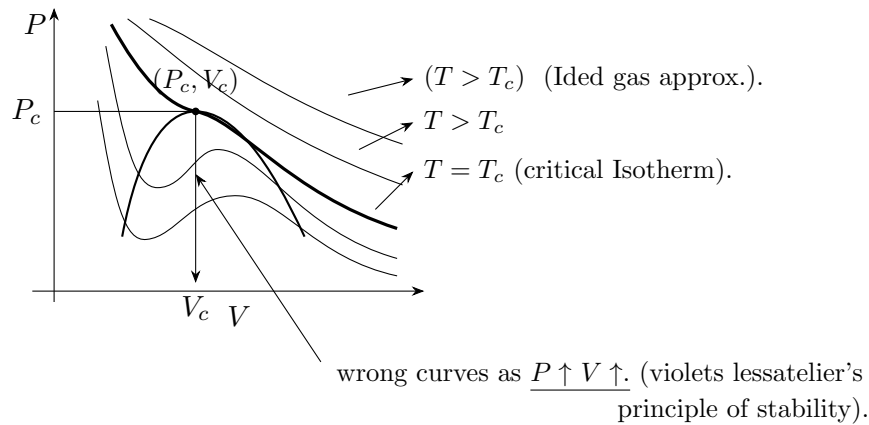


discontinuous phase transition jump in density (ρ).



Critical point is 2nd order or continuous phase transition, and coexisting curve is series of discontinuous / 1st order phase transitions.

P-V diagram :- (only for boiling curve).



$$P = \frac{nRT}{V - nb} - \frac{an^2}{V^2} \quad \text{or} \quad \left(\left(P + \frac{an^2}{V^2} \right) (V - nb) = nRT \right)$$

To find critical point (Inflection point).

$$\left. \frac{\partial P}{\partial V} \right|_{(T_c, P_c, V_c)} = 0$$

also

$$\left. \frac{\partial^2 P}{\partial V^2} \right|_{(P_c, V_c, T_c)} = 0$$

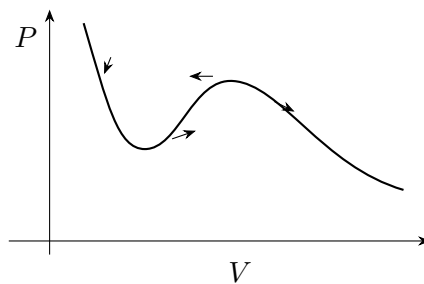
and from eqⁿ of state, we can find (P_c, V_c, T_c) .

There is no universal eqⁿ of state as eqⁿ of state depends upon interactions and interactions are hard to model universally.

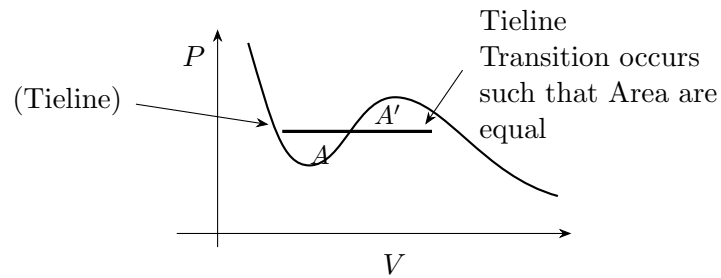
The portion where $\frac{\partial P}{\partial V}$ is positive makes compressibility

$$K_B = -\frac{1}{V} \frac{\partial V}{\partial P}$$

negative and violates thermodynamic stability. So we have to correct this portion.

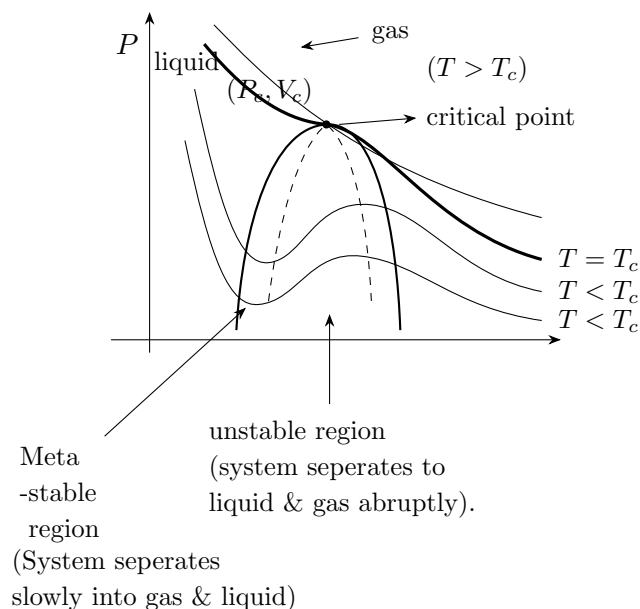


hysteresis is the problem here. So to avoid, it is observed that, $\oint P$



So some portions of allowed curve also gets omitted in observation. So vaander vaals gas eqⁿ of state is not accurate, we not only have to remove unstable region but slightly bigger region also.

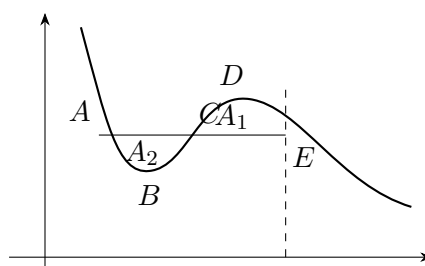
So envelope goes like this –



Any system on unstable region will instantaneously separate into some liquid & remain to gas depending upon lever rule.

Note :- In case of binary alloys (liquid alloys), the metastable portion can be made extremely slow, It may take hours, yeares to happen.

Maxwells equal Area rule :-



(maxwell's tieline construction)

(Is not quite correct though as It goes through ($\textcircled{\text{D}}$ $\rightarrow$ $\textcircled{\text{B}}$) i.e. from thermodynamic unstable region.)

let us consider $G(T, P, N) = U - TS + PV$

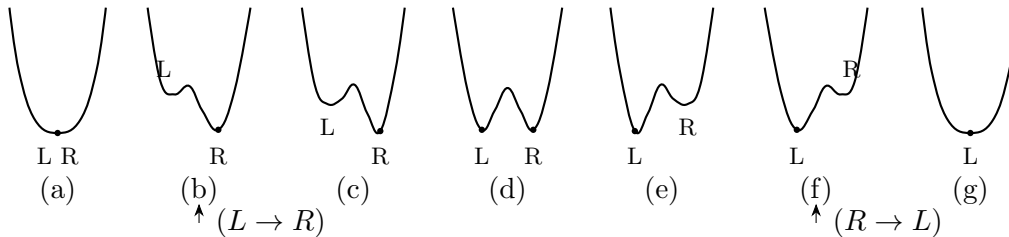
$$\begin{aligned} dG &= dU - TdS - SdT + PdV + VdP \\ &= -SdT + VdP + \mu dN \end{aligned}$$

for fixed (N, P, T) ($dG = 0$)

On an isotherm, $dG = VdP$ (as $dT = 0$).

$$\begin{aligned} G_E - G_A = 0 &= \int_A^E dG = \int_A^B VdP + \int_B^C VdP + \int_C^D VdP + \int_D^E VdP \\ &\quad \left(G = \frac{\mu}{N} \text{ should be same in gas phase \& liquid phase} \right) \\ &= \underbrace{\left(\int_A^B VdP - \int_C^B VdP \right)}_{(A_1)} + \underbrace{\left(- \int_D^C VdP + \int_D^E VdP \right)}_{(-A_2)} = 0 \\ &= \boxed{(A_1 = A_2) = A} \end{aligned}$$

Mechanical example of hysteresis (Double well potentials).



If we start at (R) i.e. right well, Transition occurs to 'L' at (f).

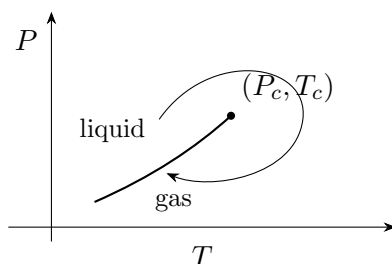
If we start at 'L' i.e. left wall (g) then transitions occur at (b). This is called hysteresis in mechanical case.

There is a branch of mathematics called catastrophe theory where people talk about change in generic potentials. (BTW) we can't have such minimum in thermal eqⁿ.

13 Lecture 28

Near critical point, many systems behave (independent of their interactions/details) identically.

We can convert from liquid to gas or gas to liquid without encountering a sharp transitions. (This is only possible because boiling curve ends at critical point).



we can't have continuous transitions from liquid to solid or vice versa. (Because this curve (melting curve) does not end. It does not end because solid has crystalline symmetry

It is symmetric under group of specific transformations like space rotations, translations. On the other hand liquid is homogeneous and isotropic, therefore liquid has much greater degree of symmetry. Solid has long range order.

order and symmetry are kind of opposites. More ordered is phase less symmetric it is; in the sense that the set of transformations under which it remains unchanged is smaller and smaller.

The most disordered state is in fact most symmetric because it looks exactly the same in every direction.

on the other hand,

liquid and gas phases are both symmetric (in every dir.).

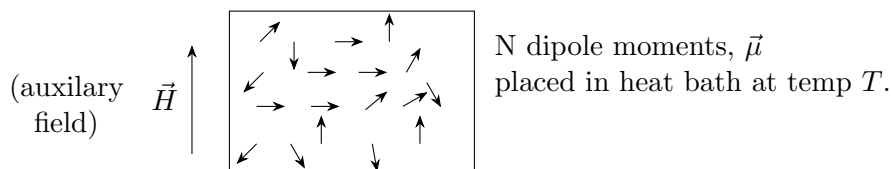
So $\exists$ a critical point

we can understand behaviour at critical point via fluid-magnet analogy.

– Model of paramagnetism

Assumption – system consists of elementary magnetic dipole moments (atomic (Spin + Angular)). oriented randomly in presence of ext. $\vec{H}_{ext}$ $\left(H = \frac{\vec{B}}{\mu_0} \right)$.

magnetic dipole moment will try to align in direction of $\vec{H}_{ext}$. So system will have some kind of symmetry.



P.E of a dipole $\vec{\mu}$ in field $\vec{H}$

$$U = -\vec{\mu} \cdot \vec{H}$$

$$U_{min} = -\mu H \quad (\theta = 0)$$

$$U_{max} = +\mu H \quad (\theta = \pi).$$

lets look at very simple model, such that

$$U = +\mu H \text{ or } -\mu H \quad (\text{one-dimension case})$$

(magnetic dipole moment of e^- can only take two values).

$$\begin{array}{ccc} E_1 & \text{-----} & (\mu H) \quad \uparrow \vec{\mu} \quad \downarrow \vec{H} \\ E_0 & \text{-----} & (-\mu H) \quad \uparrow \vec{\mu} \quad \uparrow \vec{H} \end{array}$$

lets find magnetisation (or better average Magnetisation). in $(1 - D)$

$$\vec{M} = N \text{ (Avg dipolemoments of each dipole moment).}$$

($\hookrightarrow N\langle\mu\rangle$ is only true iff all $\vec{\mu}$ are independent assuming no-dipole-dipole interactions).

(since we only want to see $\vec{M}$ along dir. of $\vec{H}$. lets drop vector)

$$\begin{aligned} M &= N \left(\frac{\mu P(\mu) + (-\mu) P(-\mu)}{P(\mu) + P(-\mu)} \right) \\ &= \frac{N (\mu e^{\beta\mu H} - \mu e^{-\beta\mu H})}{e^{\beta\mu H} + e^{-\beta\mu H}} \quad \left(\tanh x = \frac{e^x - e^{-x}}{e^x + e^{-x}} \right). \end{aligned}$$

Av. magnetisation (M) = $N\mu \tanh(\beta\mu H)$

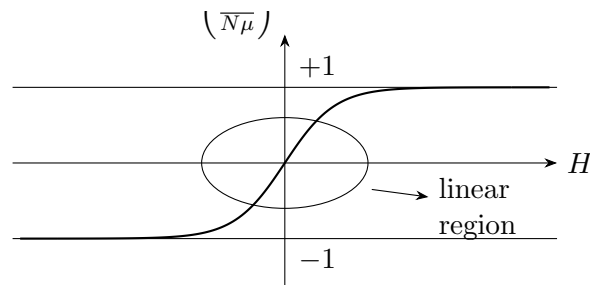
So, $\boxed{M = N\mu \tanh\left(\frac{\mu H}{k_B T}\right)}$ is magnetic eqⁿ of state.

fluid (P, V, T) of [?] (H, M) system

$$\begin{array}{cc} \uparrow & \uparrow \\ \text{field} & \text{responce} \\ \text{(Intensive)} & \text{extensive} \end{array} \quad \begin{array}{cc} \uparrow & \uparrow \\ \text{field} & \text{responce} \\ \text{(Intensive)} & \text{(extensive)} \end{array}$$

$\left(\frac{M}{N\mu}\right)$ vs H :-

$\left(\frac{M}{N\mu}\right)$ is magnetisation per atom per dipole moment.



our old defⁿ of susceptibility (χ)

$$\begin{aligned} \vec{M} &\propto \vec{H} \\ \vec{M} &= \chi \vec{H} \\ \frac{M}{H} &= \text{const} \end{aligned}$$

is not quite right as, this graph is non linear in most range of $\vec{H}$.

Isothermal susceptibility

$$\chi_T = \left(\frac{\partial M}{\partial H} \right)_{T,N} \Big|_{H=0}$$

or

(χ_T = slope at origin)

$$\chi_T = \left(\frac{\partial M}{\partial H} \right)_T \Big|_{H=0}$$

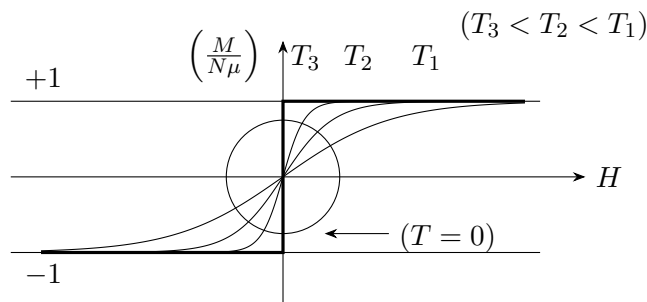
$\tanh \theta \approx \theta$ for very small θ .

$\tanh ax \approx ax$

$$= N\mu \left(\frac{\mu H}{k_B T} \right)$$

$$\left(\chi_T = \frac{N\mu^2 H}{k_B T} \right) \rightarrow \text{Curie's law (pierre curie)} \quad \left(\chi_T \propto \frac{1}{T} \right)$$

So slope at origin increases with decrease in temp.



at $T = 0$ (absolute zero)

$$\chi \rightarrow \infty$$

or slope $\rightarrow \infty$

We have neglected interaction between magnetic dipoles.

At high temperature, we can neglect dipole-dipole interaction but at very low temp, dipole-dipole interactions become large as compared to random fluctuation driven by heat bath.

All phase transitions of this kind is dependent on minimization of free energy

$$F = U - TS$$

$$U = \langle E \rangle \text{ or } \bar{E} \text{ or } \langle \mathcal{H} \rangle \quad (\text{Hamiltonian})$$

It so happens that at sufficiently low temperatures the ordering tendency of internal energy (U) overcomes effect of entropy.

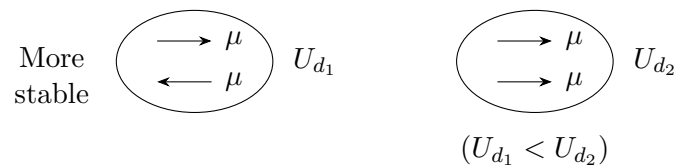
So Free energy (F) is mostly governed by U as ' T ' is very small and ' S ' (entropy) also becomes small at low temperatures ($S = k_B \ln \Omega(E)$) ($\Omega(E)$ decreases with $\underline{T}$).

So at low temp, ‘ F ’ is governed by U and it is ordered phase, and at high temp ‘ F ’ is governed by ‘ TS ’ which is very large compared to U , and this is disordered phase.

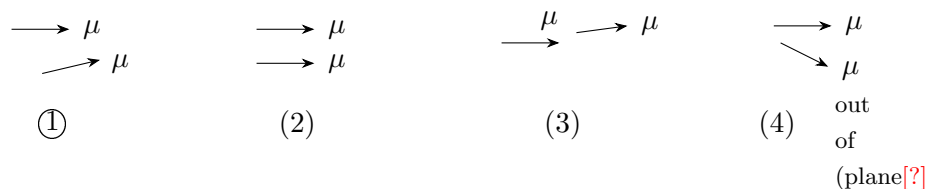
- “This is the main reason low temp is ordered state and high temp is disordered state.”

If dipole–dipole interaction is taken in account our plot or relation $\left(M = N\mu \tanh\left(\frac{\mu H}{k_B T}\right) \right)$ fails.

For two dipoles, if U_d represents interaction energy



dipole–dipole interaction is not isotropic, below configuration does not have same energy.



If we rely on dipole–dipole interactions alone, we won’t have bar magnet at all. Because one ($\uparrow \mu$) will flip another in reverse direction ($\downarrow \mu$), so that the net dipole moment cancels out & magnetisation becomes zero.

$$(\uparrow \downarrow \uparrow \downarrow \uparrow \downarrow \uparrow \downarrow \cdots).$$

net ($M = 0$).

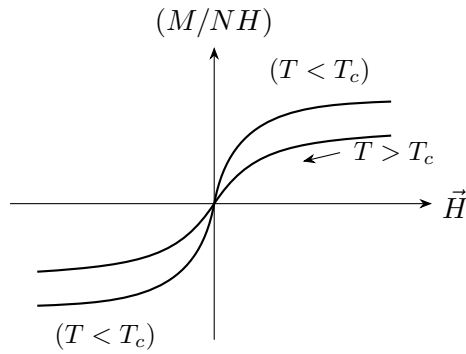
But we do see permanent magnets, so the phenomena of permanent magnetism is more subtle than this and is not connected to dipole–dipole interactions, It is connected to a quantum mechanical effect called the exchange Interaction which actually favours ($\uparrow \uparrow$) over ($\uparrow \downarrow$). and it in fact overcomes dipole–dipole interactions. ($\uparrow \downarrow$).

Exchange Interaction is very strong short range interaction It decays exponentially. (where as dipole–dipole inter-action is long range $\left(\frac{2K(\vec{p} \cdot \hat{r})}{r^3} \right)$ & weak).

There are magnets where dipole–dipole interaction becomes more important/dominant and it leads to Anti-ferromagnetism : ($\uparrow \downarrow \uparrow \downarrow \uparrow \downarrow \uparrow \downarrow$).

If we take exchange interactions in account we see that the curve b/w $\left(\frac{M}{NH} \right)$ vs $\vec{H}$, attains infinite slope at finite temp. ($T > 0K$) instead of ($T = 0K$).

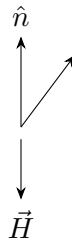
Below curie temp (T_c) the curve attains infinite slope at origin.



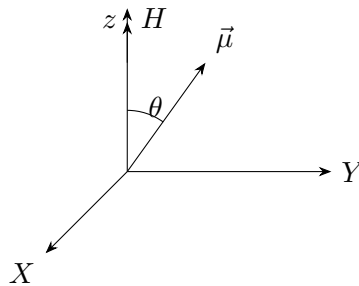
So this is discontinuous phase transitions at (T_c) curie temp.

This phase transition is from paramagnetic phase to ferromagnetic phase (M is very high even at $H = 0$).

For general case $(E = -\mu H \cos \theta)$ (3-D case)



lets compute the magnetisation.



– Also assuming no dipole-dipole interaction.

$$M = N \langle \mu \rangle$$

(our magnetisation)

$$\langle \mu \rangle = \frac{\int \mu \cos \theta P(\theta) d\Omega}{\int P(\theta) d\Omega} = \frac{\int \mu \cos \theta e^{+\beta \mu H \cos \theta} d\Omega}{\int e^{+\beta \mu H \cos \theta} d\Omega}$$

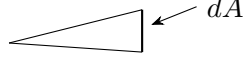
($d\Omega$ = solid angle)

Here we have ignored K.E, Rotational energy of these dipole moments, we have only used magnetic energy.

or

$$\int d\Omega = \int_{-1}^1 d(\cos \theta) \int_0^{2\pi} d\varphi$$

$$\int d\Omega = \int_0^\pi \sin \theta d\theta \int_0^{2\pi} d\varphi$$



$$\begin{aligned} d\Omega &= \frac{dA}{r^2} \\ &= \frac{(r d\theta)(r \sin \theta d\varphi)}{r^2} \end{aligned}$$

$$d\Omega = \sin \theta d\theta d\varphi$$

$$\int d\Omega = \iint \sin \theta d\theta d\varphi$$

$$M = N \langle \mu \rangle$$

$$= N\mu \frac{\int \cos \theta e^{\beta\mu H \cos \theta} \sin \theta d\theta \int_0^{2\pi} d\varphi}{\int_0^\pi \sin \theta d\theta \int_0^{2\pi} d\varphi}$$

$$(\cos \theta = t, \quad -\sin \theta d\theta = dt)$$

$$= \frac{N\mu (2\pi) (-) \int_1^{-1} t e^{\beta\mu H t} dt}{2\pi \int_0^\pi e^{\beta\mu H \cos \theta} \sin \theta d\theta}$$

$$= \frac{2\pi N\mu \int_{-1}^1 t e^{\beta\mu H t} dt}{2\pi \int_{-1}^1 e^{\beta\mu H t} dt}$$

$$= \frac{N\mu \left(\frac{t e^{\beta\mu H t}}{(\beta\mu H)} - \frac{e^{\beta\mu H t}}{\beta^2 \mu^2 H^2} \right) \Big|_{-1}^1}{\left(\frac{1}{\beta\mu H} \right) (e^{\beta\mu H}) \Big|_{-1}^1}$$

$$= \frac{N\mu^2 H \beta \left(e^{\beta\mu H} \left(\frac{1}{\beta\mu H} - \frac{1}{\beta^2 \mu^2 H^2} \right) - e^{-\beta\mu H} \left(\left(\frac{-1}{\beta\mu H} \right) - \frac{1}{\beta^2 \mu^2 H^2} \right) \right)}{(e^{\beta\mu H} - e^{-\beta\mu H})}$$

$$= N\mu \left[\frac{e^{\beta\mu H} + e^{-\beta\mu H}}{e^{\beta\mu H} - e^{-\beta\mu H}} - \frac{1}{\beta\mu H} \right]$$

$$M = N\mu \left[\coth(\beta\mu H) - \frac{1}{\beta\mu H} \right]$$

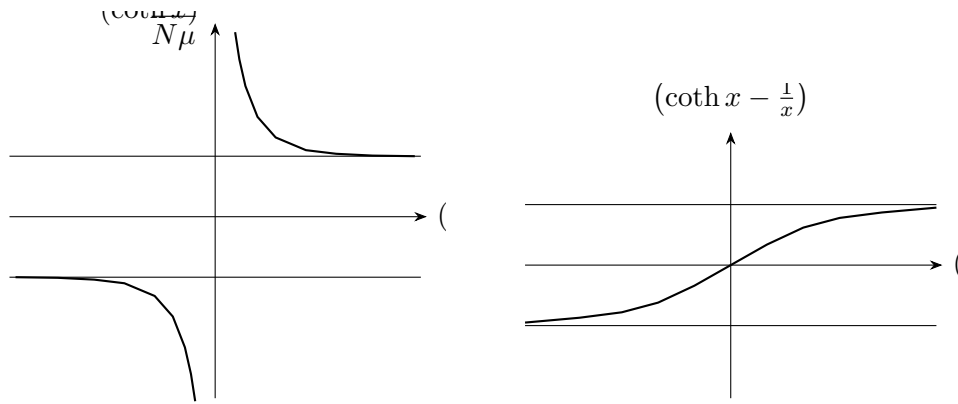
Previously when we only assumed ($E = \pm\mu H$)

$$M = N\mu \tanh x \quad x = \beta\mu H$$

↓ Now when we used all possible orientations

$$M = N\mu \left(\coth x - \frac{1}{x} \right) \quad = \text{Langevin functions.}$$

Let us plot $\left(\frac{M}{N\mu} \right)$ vs $\left(\coth x - \frac{1}{x} \right)$ vs (x)



$$\begin{aligned} \coth x &= \frac{\cosh x}{\sinh x} = \frac{\left(1 + \frac{x^2}{2} + \dots\right)}{x \left(1 + \frac{x^2}{6} + \dots\right)} \\ &= \left(\frac{1}{x}\right) \cdot \left(1 + \frac{x^2}{2} + \dots\right) \left(1 - \frac{x^2}{6} + \dots\right) \\ &= \frac{1}{x} \left(1 + x^2 \left(\frac{1}{2} - \frac{1}{6}\right) + \dots\right) \\ &= \frac{1}{x} \left(1 + \frac{2x^2}{6}\right) \\ \coth x &\approx \frac{1}{x} + \frac{x}{3} \end{aligned}$$

So $\left(\coth x - \frac{1}{x} \right) \approx \left(\frac{x}{3} \right)$

So,

$$\frac{M}{N\mu} = \left(\coth \frac{\mu H}{k_B T} - \frac{k_B T}{\mu H} \right)$$

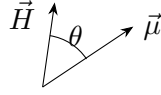
So, near origin ($H = 0$)

$$\left. \frac{M}{N\mu} \right|_{H \rightarrow 0} \approx \frac{\mu H}{3k_B T}$$

So $\left(\chi_T = \frac{N\mu^2}{3k_B T}\right)$ (χ_T = Isothermal susceptibility)

→ No of dimensions is '3' in our case

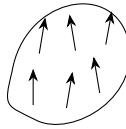
(For 2-D) :- μ is allowed to have any dirn. in a plane



$$M = N\langle\mu\rangle$$

$$M = N \frac{\int_0^{2\pi} \mu \cos \theta e^{+\beta\mu H \cos \theta} d\theta}{\int_0^{2\pi} e^{\beta\mu H \cos \theta} d\theta}$$

$$\frac{M}{N\mu} = \frac{\int_0^{2\pi} \cos \theta e^{\beta\mu H \cos \theta} d\theta}{\int_0^{2\pi} e^{\beta\mu H \cos \theta} d\theta} \quad \leftarrow \text{(Bessels function } J_0)$$



$$\frac{M}{N\mu} = \frac{\int \cos \theta \left(1 + \frac{\beta\mu H \cos \theta}{1} + \frac{(\beta\mu H \cos \theta)^2}{2!} + \dots\right) d\theta}{\int_0^{2\pi} \left(1 + \beta\mu H \cos \theta + \frac{(\beta\mu H \cos \theta)^2}{2!} + \dots\right) d\theta}$$

We are only interested in finding (χ_T) i.e. slope at origin. So, let's expand $(e^{\beta\mu H \cos \theta})$ to first order.

$$\begin{aligned} \frac{M}{N\mu} &= \frac{0 + \beta\mu H \int_0^{2\pi} \cos^2 \theta d\theta}{2\pi + \beta\mu H \left(\underbrace{\int_0^\pi \cos \theta d\theta}_{\rightarrow 0}\right)} = \frac{\beta\mu H \pi}{2\pi} \\ &= \frac{\beta\mu H}{2} \end{aligned}$$

So slope

$$\left.\frac{\partial M}{\partial H}\right|_{H \rightarrow 0} = \frac{N\beta\mu^2}{2} = \frac{\mu^2 N}{(2k_B T)} \quad \rightarrow \text{dimension is 2.}$$

$$\chi_T = \frac{N\mu^2}{2k_B T}$$

what we saw that ($\chi_T \propto \frac{1}{T}$) but it is not valid at small temp ($T \rightarrow 0$ or $T = 0$). So we need to fix it.

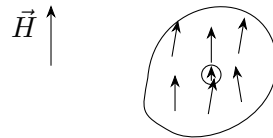
Just like to go from ideal gas to real gas (use interactions) we used the van der Waals gas eqn, similarly here we use Weiss molecular field theory.

Weiss molecular field theory

The idea is that in the ferromagnetic medium each dipole moment experience not only the ext. field $\vec{H}$ but also the internal mag. field due to other dipole moments.

$$H \xrightarrow{\text{replace}} H_{\text{eff}} = H + \lambda \frac{M}{N}$$

↓ to make intensive quantity to add to $\underline{H}$.



$$H_{\text{int}} \propto M \qquad H_{\text{int}} = \lambda M$$

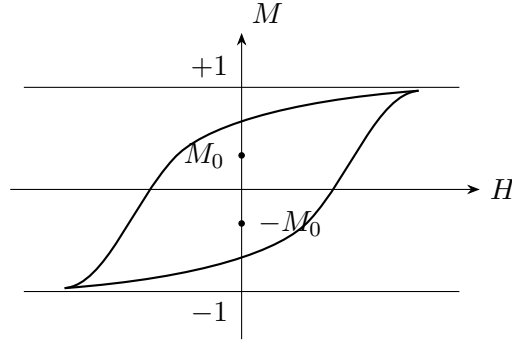
$$M = N\mu \tanh\left(\frac{\mu\left(H + \frac{\lambda M}{N}\right)}{k_B T}\right) \quad (\text{for 1-D}).$$

This is transcendental equation, we can only solve it by Numerical methods.

Let us see its behaviour near $\underline{H = 0}$.

$$M_0 = N\mu \tanh\left(\frac{\mu\lambda M_0}{Nk_B T}\right) \quad \left(\begin{array}{l} M_0 = \text{magnetisation} \\ \text{per unit} \\ \text{dipole moments} \end{array} \right)$$

- 1) $M_0 = 0$ is always a solution (Paramagnetic solution). ($H = 0$, $M = 0$)
- 2) But ferromagnet have hysteresis.



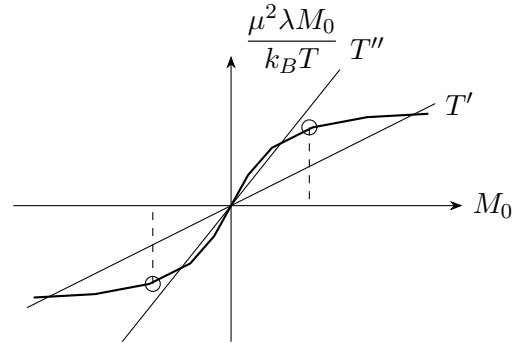
Near $M \approx 0$

$$M_0 = N\mu \left(\frac{\mu\lambda M_0}{Nk_B T} \right)$$

$$\approx \frac{N\mu^2\lambda M_0}{Nk_B T} \approx \frac{\mu^2\lambda M_0}{k_B T}$$

Lets plot $\left(\frac{\mu^2\lambda M_0}{k_B T} \right)$ vs (M_0)

(at high temp $T' > T_0$, slope is less), (for smaller temp T'' slope is even higher).



Intersection happens if slope > 1

$$\frac{\mu^2\lambda}{k_B T} > 1 \quad \text{or} \quad \boxed{T < \frac{\mu^2\lambda}{k_B}}$$

So $\left(T = \frac{\mu^2\lambda}{k_B} \right)$ is critical point.

The negative solution for M_0 is unstable; so we only have one positive root of M_0 . Which leads to ferromagnetism.

So below a certain critical temp, the cooperative tendency of all these magnetic moments to align in same dir. will dominate over disrupting tendency of entropy (S) and we have order.

$$F = U - TS \quad (U \gg TS)$$

at low temp (T) $F \approx U$ (ordered phase) (ferromagnetic)

But above (T_c) disorder wins and interaction is not strong enough to maintain order.

$$F = U - TS$$

$$(TS \gg U)$$

This is how phase transition occurs b/w paramagnet and ferromagnet. Now we will see how it is similar to case of fluids.

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We can also have ferroelectric transitions of permanent electric dipole moments, which are accompanied by structural phase transitions (crystalline phase transitions which would lead to shapes of unit cells which have permanent electric dipole moments.

The behaviour near the critical region of eqn of state

$$M = N\mu \tanh \left(\frac{\mu \left(H + \frac{\lambda M}{N} \right)}{k_B T} \right)$$

would be very similar to that of fluid near its

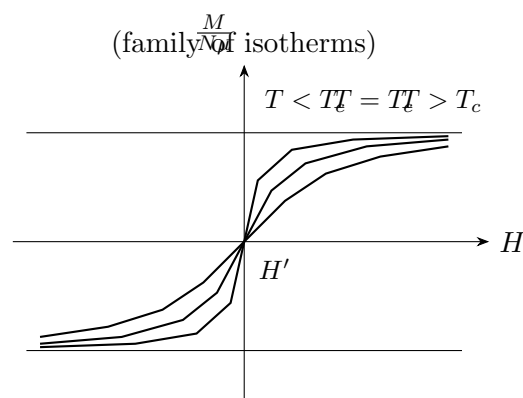
critical region. We would like to find out the non zero magnetisation (M) for $H = 0$.

$$M_0 = N\mu \tanh \left(\frac{\mu \lambda M_0}{N k_B T} \right)$$

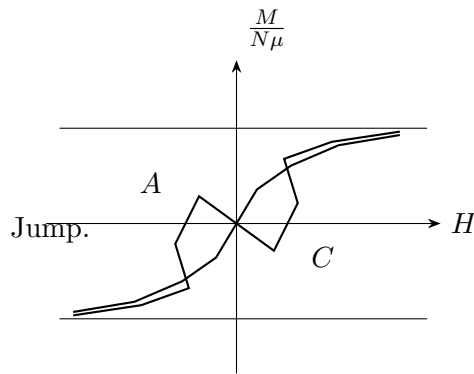
(Digression) also our model $M = N\mu \tanh \left(\frac{\mu \left(H + \frac{\lambda M}{N} \right)}{k_B T} \right)$ does not include hysteresis phenomena.

Hysteresis actually arose from presence of multiple domains in magnetic material. These domains align in right direction in presence of strong $\vec{H}$. For a single domain our model works perfectly as a single domain can't show hysteresis.

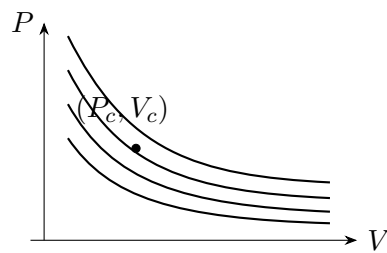
Family of Isotherms :-



$$\left(\text{Critical Isotherm} \quad \frac{\partial M}{\partial H} = 0 \ \& \ \frac{\partial^2 M}{\partial H^2} = 0 \right)$$



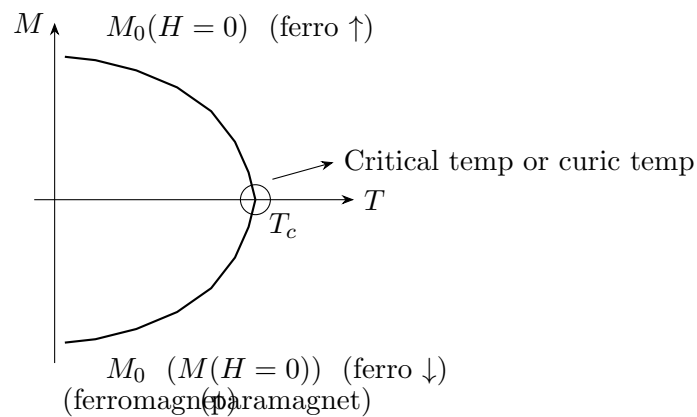
curve $A \rightarrow C$ is unphysical as ($H \uparrow$ in $-ve$ direction & $M \uparrow$ in $+ve$ direction).



← Isotherms in case of fluid.

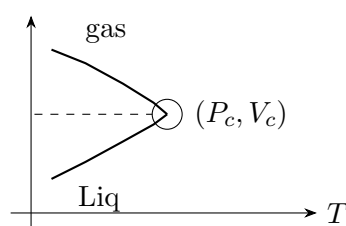
$$\rightarrow \left(\text{Critical Isotherm } \frac{\partial P}{\partial V} \bigg|_{(P_c, V_c)} = 0, \text{ \& } \frac{\partial^2 P}{\partial V^2} \bigg|_{(P_c, V_c)} = 0 \right)$$

let us plot (M vs T)



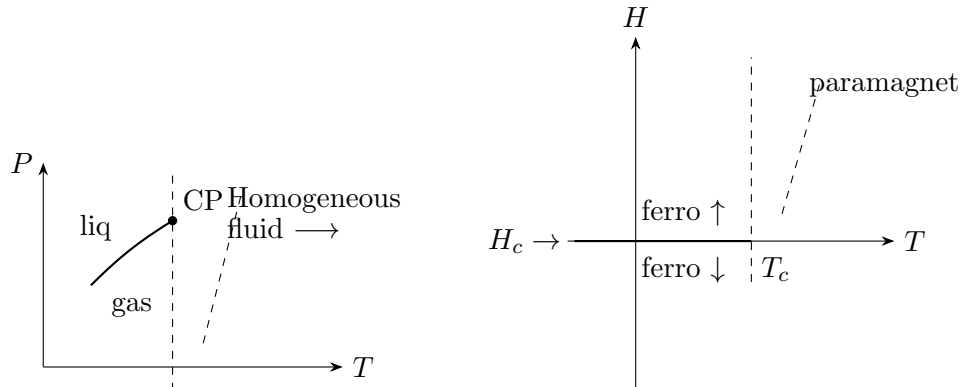
In case of (magnet)

In case of fluid



→ discontinuous transitions

→ parabolic region

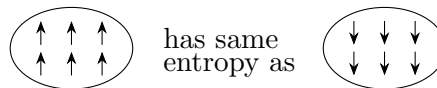


$$\frac{dP}{dT} = \frac{L}{T\Delta V} = \frac{T\Delta S}{T\Delta V} = \frac{\Delta S}{\Delta V} = \frac{S_{gas} - S_{liq}}{v_{gas} - v_{liq}}$$

↓ specific volume

$$\frac{dH}{dT} = \frac{\Delta S}{\Delta M}$$

In case of fluid as well as f. magnet, we have one line ending at critical pt. [?] from ferro ↑ to ferro ↓ [?] with ΔS [?]



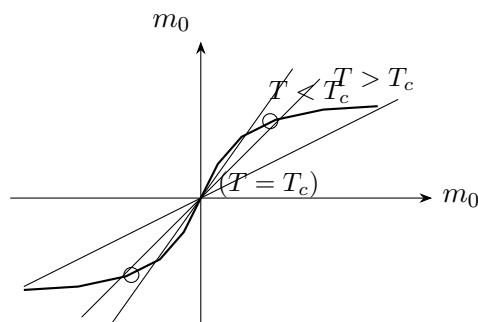
$$\left(\frac{dH}{dT} = \frac{\Delta S}{\Delta M} = \frac{0}{\text{nonzero}} = 0 \right) \quad \checkmark$$

let's find

$$M_0 = N\mu \tanh\left(\frac{\mu\lambda M_0}{Nk_B T}\right)$$

$$\text{let } \left(\frac{M_0}{N\mu} = m_0\right)$$

$$m_0 = \tanh\left(\frac{\mu^2 \lambda m_0}{k_B T}\right)$$



T_c is given by when slope is 45°

$$\left(\frac{\mu^2 \lambda}{k_B T} = 1 \right)$$

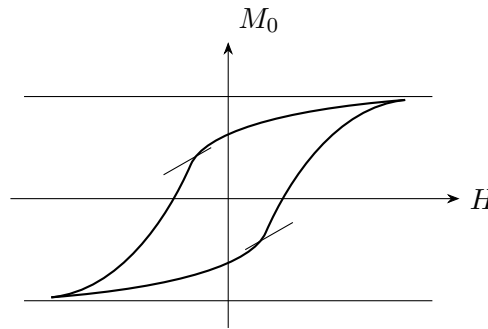
$$\left(T_c = \frac{\mu^2 \lambda}{k_B} \right).$$

$(M_0 = 0)$ root is not stable for $(T < T_c)$ as magnet becomes ferromagnet instead of paramagnet.
We can show it using $F = U - TS$ $(T < T_c)$ $(F \approx U) \rightarrow$ ordered phase

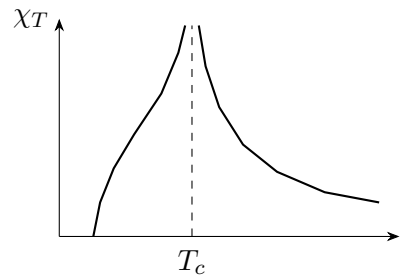
Susceptibility at $(H = 0)$.

$$\chi_T = \left(\frac{\partial M}{\partial H} \right)_T \Big|_{H=0}$$

even though (M) is discontinuous we can define χ_T as slope are continuous.



Lets plot $(\chi_T$ vs T)



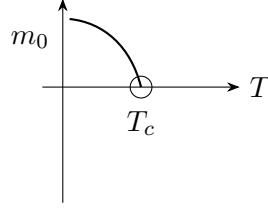
$$m_0 = \tanh \left(\frac{\mu^2 \lambda m_0}{k_B T} \right) = \tanh \left(\frac{T_c m_0}{T} \right)$$

at $T = T_c$ $m_0 = \tanh m_0 \Rightarrow (m_0 = 0)$ is only solution (paramagnet).

but below $(T < T_c)$ we have other solutions.

let us expand $\tanh x$, near $[?]$

$$\tanh x = \frac{\sinh x}{\cosh x} = \frac{e^x - e^{-x}}{e^x + e^{-x}} = \frac{\left(1 + x + \frac{x^2}{2!} + \dots \right) - \left(1 - x + \frac{x^2}{2!} - \frac{x^3}{3!} + \dots \right)}{\left(1 + x + \frac{x^2}{2!} + \frac{x^3}{3!} \right) + \left(1 - x + \frac{x^2}{2!} - \frac{x^3}{3!} + \dots \right)}$$



$$= \frac{2 \left(x + \frac{x^3}{3!} \right)}{2 \left(1 + \frac{x^2}{2!} + \frac{x^4}{4!} + \dots \right)} = \left(x + \frac{x^3}{3!} + \dots \right) \left(1 - \frac{x^2}{2!} - \dots \right)$$

$$= \left(x - \frac{x^3}{2!} + \frac{x^3}{3!} - \frac{x^5}{3! 2!} + \dots \right)$$

$$\tanh x \approx x + x^3 \left(\frac{1}{6} - \frac{1}{2} \right) = x - \frac{x^3}{3} + \dots$$

$$m_0 = \tanh \left(\frac{T_c}{T} m_0 \right)$$

$$m_0 \approx \frac{T_c}{T} m_0 - \frac{1}{3} \left(\frac{T_c}{T} m_0 \right)^3$$

for m_0 other than zero,

$$m_0 \left(1 - \frac{T_c}{T} \right) \approx -\frac{1}{3} \left(\frac{T_c}{T} m_0 \right)^3$$

$$1 - \frac{T_c}{T} \approx -\frac{1}{3} \left(\frac{T_c}{T} \right)^3 m_0^2$$

$$\left(\frac{T_c - T}{T} \right) \approx \frac{1}{3} \left(\frac{T_c}{T} \right)^3 m_0^2$$

($T < T_c$) if T is very near to T_c ($\frac{T_c}{T} \approx 1$)

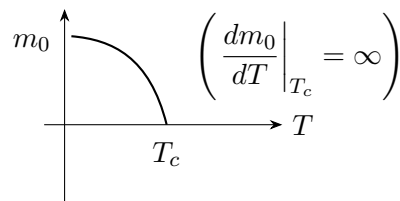
$$\Rightarrow \boxed{m_0 \sim \pm \sqrt{T_c - T}}$$

$m_0 \approx \pm (T_c - T)^{1/2} \rightarrow$ mean field critical exponent (β)

here ($\beta \neq \frac{1}{k_B T}$) (Just a symbol)

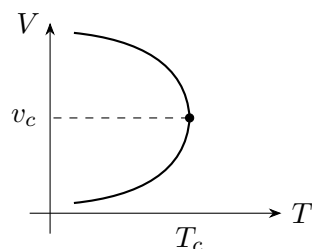
$$\left. \frac{dm_0}{dT} \right|_{T=T_c} = \frac{(-1)}{\pm 2\sqrt{T_c - T}} \rightarrow \infty.$$

So behaviour of m_0 near T_c is



Exactly same relation arose in case of fluid.

$$\left((v - v_c) \sim \pm \sqrt{T_c - T} \right)$$



(H.W) show that $(v - v_c) \sim \pm \sqrt{T_c - T}$.

we can see that (in both cases critical exponent is $(\frac{1}{2})$) which arose because (both models are examples of mean field theory models. i.e. In van der vaals case we assume $P \neq \frac{RT}{v-b}$ but $P = \frac{RT}{v-b} - \frac{a}{v^2}$ because of attraction of each molecules. In same way

here we assume every magnetic moment another magnetic moment and internal $\vec{H}$ is proportional to magnetisation itself, we have exactly the same philosophy and that is what led to critical exponent.

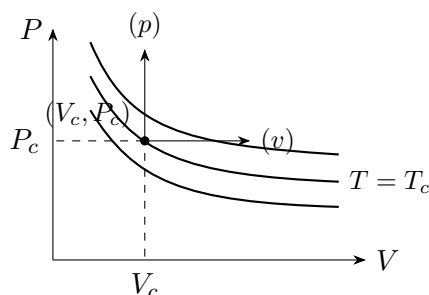
(Q) Does experiments agree with critical exponent to be $(1/2)$ or how accurately we can get closer to T_c and controll temp. to see it.

Ans $\sim$ In most cases we essentially see exponent closer to $1/2$ but, but, if we do careful experiments near T_c we do not see half at all. we see $(1/3$ or $0.3)$ for it we have to get very-very close to T_c to see it.

Temp is one of hardest thing to controll, best we can controll is of (milli-degree) of accuracy.

Frequency is most accurately measurable quantity, we can measure frequency in one part in 10^{15} . There are astronomical objects/clocks (pulsars) whose rate of change of time period can be computed to one part in 10^{19} .

(Q) How does critical Isotherm behave? (near (V_c, P_c)).



shifting origin to see behaviour near (V_c, P_c)

$$\left(\frac{\partial P}{\partial V} \right)_{T_c} = \left(\frac{\partial^2 P}{\partial V^2} \right)_{T_c} = 0$$

$$p = \frac{P - P_c}{P_c} \quad v = \frac{V - V_c}{V_c}$$

The curve near (V_c, P_c) is like

$$(p \sim |v|^\delta) \longrightarrow \text{mean field critical exponent } (\delta)$$

or $(|v| \sim p^{1/3})$

lets look at critical isotherm for ferro magnetic transitions

$$M = N\mu \tanh \left(\frac{\mu \left(H + \frac{\lambda M}{N} \right)}{k_B T} \right)$$

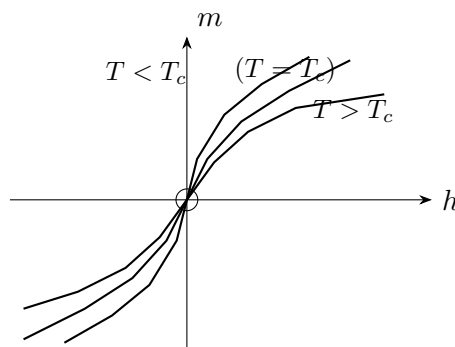
$$\text{let } m = \frac{M}{N\mu}$$

$$\begin{aligned} m &= \tanh \left(\frac{\mu}{k_B T} (H + \lambda m \mu) \right) \\ &= \tanh \left(\frac{\lambda \mu^2}{k_B T} \left(\frac{H}{\lambda \mu} + m \right) \right) \\ &= \tanh \left(\frac{T_c}{T} (h + m) \right) \end{aligned}$$

$$\text{let } \left(\frac{H}{\lambda \mu} = h \right)$$

for critical isotherm $(T = T_c)$

$$m = \tanh(h + m)$$



$\longrightarrow$ critical isotherm

at $(h = 0), (m = 0)$ m is zero. (for $T > T_c$)

$$m = \tanh(h + m) \approx (h + m) - \frac{(h + m)^3}{3}$$

$$\Rightarrow (m + h)^3 \approx h$$

h is much smaller (near origin) so $m^3 \sim h$

$$\boxed{m \sim h^{1/3}} \quad (h \text{ is field variable})$$

Just like $(|v| \sim p^{1/3})$ (p is field variable)

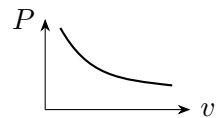
(m, v) are response variables.

Similarly, χ_T and compressibility behave exactly same near critical point.

$$\chi_T = \left(\frac{\partial M}{\partial H} \right)_T \bigg|_{H=0} ;$$

$$k_T = \frac{-1}{v} \frac{\partial v}{\partial P}$$

$$\frac{\partial P}{\partial v} = 0 \text{ at critical point}$$



$\left(\begin{array}{l} \text{curie} \\ \text{weiss law} \end{array} \right) \Leftarrow$

$$\chi_T = \frac{1}{|T - T_c|}$$

$$(\chi_T|_{T=T_c} = \infty) \longleftrightarrow (k_T = \infty)$$

$$\chi_T = \frac{1}{|T - T_c|^\gamma} \longrightarrow \gamma \text{ critical point}$$

$(\gamma \neq C_P/C_V)$ just a symbol.

Experimental values of $(\gamma \sim 1.3)$.

All these phase transitions can be explained by Landau's theory ((1937). (Idea of broken symmetry leads to such phase transitions of all kinds of systems).